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Disentangling Moral Hazard and Adverse Selection†

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While many real-world principal-agent problems have both moral hazard and adverse selection, existing tools largely analyze only one at a time. Do the insights from the separate analyses survive when the frictions are combined? We develop a simple method—decoupling—to study both problems at once. When decoupling works, everything we know from the separate analyses carries over, but interesting interactions also arise. We provide simple tests for whether decoupling is valid. We develop and numerically implement an algorithm to calculate the decoupled solution and check its validity. We also provide primitives for decoupling to work and analyze several extensions. (*JEL* D82, D86)

Settings with both screening and moral hazard are ubiquitous. A firm wants workers to self-identify as more or less able and to tailor their incentives and effort accordingly. An insurance company faces a trade-off between risk sharing and the customer's incentive to exercise care and tailors that trade-off to the privately known riskiness of the customer. An investor wants an entrepreneur to both reveal what he knows about the quality of the project and choose an appropriate level of effort. A social planner facing workers with different abilities designs a tax code that trades off equity concerns and effort incentives.

While much is known about each adverse selection and moral hazard separately, comparatively little is known when the two are present at once, especially when the agent is risk averse. How a given type is optimally motivated may depend on the details of who wants to imitate that type, so the powerful characterization we have for moral hazard may lose its relevance. The ability to screen might depend in complicated ways on different types' incentives to deviate in their choice of effort, potentially obviating what we know about adverse selection problems. Hence, little of what we know about the separate problems need be relevant when

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the two problems are intertwined. Nor may such a model be tractable except in very simplified cases.[1]

The main message of this paper is that things are often better than they might seem. In many settings, the design of an optimal menu of contracts "disentangles." First, for each menu item, the compensation scheme is in fact of the standard form that is well understood from the moral hazard setting (see Holmström 1979; Mirrlees 1975; and the enormous literature that follows). Thus, everything we know from that setting continues to be relevant. Second, the screening aspect of the problem can in a strong sense be thought about separately from the moral hazard problem, and so the optimal menu of contracts shares many of the properties of the solution to a standard screening problem (see Maskin and Riley 1984; Guesnerie and Laffont 1984; and a vast associated literature).

Disentangling has four main benefits. First, we get clear economic insight into an otherwise opaque setting. Second, we recover a substantial degree of tractability that is both of analytical value and opens the door to numerical exploration in applied settings. Third, we provide a justification for much of the literature that has studied moral hazard or adverse selection separately in settings where in reality both are clearly present. While studying the two issues simultaneously adds important nuance, our results establish that much of the insight of those separate studies survives. Finally, because the setting is tractable, we can identify interesting interactions between the two informational frictions. In sum, we provide a workhorse for studying moral hazard and adverse selection in the myriad settings where both are present.

To achieve these goals, we analyze a canonical principal-agent model. A risk-averse agent has a hidden type and takes a hidden effort, where higher types are more capable, with lower disutility and marginal disutility of effort. A mechanism recommends an effort for each announced type and compensates the agent based on his announced type and the realization of a signal that depends stochastically on his effort. Thus, the agent, by announcing his type, is effectively choosing over a menu of compensation schemes. The principal maximizes her expected profit subject to the willingness of the agent to participate, reveal his type, and take the recommended effort facing the resulting compensation scheme.

For each informational friction alone, the set of deviations for the agent is one-dimensional. Hence, in the adverse selection case, Mirrlees (1975) and Myerson (1981) give us a sweeping incentive compatibility characterization, while in the moral hazard case, the first-order approach (Rogerson 1985; Jewitt 1988) drastically simplifies the incentive constraints. But here, an agent can "double deviate" by first misrepresenting his type and then choosing an effort level other than the one recommended for the type announced. There is thus a double continuum of possible deviations, and there is no known analogous simplification other than those that exist in the single-friction cases. The main purpose of this paper is to provide and explore

[1] For adverse selection, see, for example, Guesnerie and Laffont (1984) and the textbook treatment in chapter 7 of Fudenberg and Tirole (1991). For moral hazard, see the seminal papers by Holmström (1979) and Grossman and Hart (1983). See also the textbook treatments in Laffont and Martimort (2001) and Bolton and Dewatripont (2005), which present examples with both moral hazard and adverse selection with either two types or risk neutrality.

such a simplification and to illustrate its usefulness in applications both theoretical and numerical.

We begin by considering a significantly relaxed problem in which only two local incentive constraints for the agent are considered. The first of these is that the agent should not want to locally deviate in his action given a truthful report of his type. The second is that he should not want to locally change his announcement of type given that he plans to follow the recommended action for the type he announces. To analyze this problem, we argue first that because only the first-order condition for effort is considered, in any optimal solution, the compensation scheme for each type is of the standard Holmström-Mirrlees form for the moral hazard problem. The second constraint, which determines the utility of each type, has very much the flavor of a standard relaxed screening problem.

We thus solve the relaxed problem by a comparatively simple four-step procedure, *decoupling*. First, using standard tools from the moral hazard literature, we find the principal's least-cost way to implement any given effort for any given type of the agent and any given level of utility that he must receive. This gives us a cost function that depends on effort, type, and utility. Second, we use this cost function as an input to a screening problem that is nonlinear in the agent's utility level but is still tractable.[2] Third, using the effort and utility levels yielded by this screening problem, we substitute back in the appropriate Holmström-Mirrlees contract for each type to arrive at a candidate solution to the original problem. Finally, we provide simple and tractable conditions to check whether this candidate solution is in fact feasible, and hence optimal, in the problem with the full set of incentive-compatibility constraints.

Because the relaxed program that underlies decoupling is so much simpler than the full mechanism design problem, it is numerically tractable. We develop and implement an efficient algorithm that takes a specification of the primitives of the model—the information structure, distribution over types, principal's benefit of effort, utility function of the agent, and the agent's cost of effort—and numerically solves for the candidate solution. As we will see, checking the feasibility of the candidate solution is also numerically straightforward. The algorithm allows the applied modeler to take the setting under consideration and easily explore the properties of the optimal solution and how they vary with the parameters.

When decoupling is valid, the implications of our analysis go beyond what is known about each problem separately. We show that satisfying the double-deviation incentive constraints requires a strictly positive lower bound on how fast the action increases as a function of the agent's type. This is in contrast to the classic screening setting where only weak monotonicity is needed. There are also interesting interactions between the distortions that each informational friction implies separately and those of the joint problem. We use our algorithm to explore the ways in which the two information frictions interact in examples, and contrast the optimal menu with those that arise in the cases with either one or neither of the informational frictions.

When does decoupling work? As the marginal disutility of effort to the agent falls with his type but rises with effort, the marginal cost of effort at the optimum could

[2] Indeed, we provide a new existence proof of a solution for this screening problem without quasilinear utility.

rise or fall with the ability of the agent. We show that a sufficient condition for feasibility of the decoupled solution is the *increasing marginal cost* condition (*IMC*) that the recommended effort in the solution to the screening problem rises fast enough so that a more capable agent faces a higher marginal disutility of effort.

Condition *IMC* is on an endogenous object, the recommended effort function. But *IMC* allows us to explore the validity of decoupling in two distinct but complementary ways. First, *IMC* is trivial to check in numerical solutions to the decoupled problem. Hence, an applied modeler can solve their preferred model and simply check whether the solution satisfies the requisite condition. Our numerical exploration suggests that this will often—but not always—be the case.

Second, we explore primitives under which the solution to the decoupling program is guaranteed to satisfy *IMC*. We derive two such sets of primitives. In the first, we consider the *linear output* case in which the cumulative distribution at each signal changes linearly with effort. This class is often used in economic applications, particularly when the signal is binary (for example, success or failure, accident or no accident). Our second set of conditions shows that decoupling works if the agent's outside option is sufficiently high.[3] In each case, we provide conditions on the agent's utility over income, disutility of effort, and distribution of types under which *IMC* holds.

We close with six extensions of our results. First, we observe that in the linear output case, *IMC* is not just sufficient but also necessary for feasibility. This allows us to analyze the linear case even if decoupling fails. Second, we provide an alternative sufficient condition, the *single crossing condition* (*SCC*), which is also sufficient for a solution to the decoupled problem to be feasible and hence solve the original problem. Third, we use *SCC* to extend our analysis to a setting with common values. Fourth, we examine optimal exclusion both theoretically and numerically. Fifth, we show that whenever the decoupling is valid, a deterministic contract is optimal even if one allows the principal to randomize. Finally, we study the problem of a utilitarian social planner who maximizes a weighted average of firm profits and consumers' welfare.

The literature on optimal contracts with adverse selection, moral hazard, and a risk-averse agent is small. Faynzilberg and Kumar (1997) analyze a model where the agent's type enters solely into the signal distribution and examine a relaxed problem that only considers the local incentive constraints under a separability condition on the signal distribution. Given the reparameterization we discuss in Section VIII, we subsume their model. Baron and Besanko (1987) analyze a purchaser facing a supplier who has private information about their costs and takes an unobservable action. They derive necessary properties of optimal contracts given local constraints. Neither of these papers takes advantage of the tractability that decoupling brings to the analysis. Fagart (2002) studies the same combination of moral hazard and adverse selection as us and discusses decoupling and some of its implications. None of these papers tackles the crucial issue of when a solution to the first-order conditions is globally incentive compatible, nor of what primitives ensure that decoupling

[3] The result is thus most relevant when having the agent participate is sufficiently important to the principal, as, for example, if the price of output is sufficiently high or the firm being run is large.

yields a solution that satisfies these conditions.[4] Gottlieb and Moreira (2013) analyze a problem with two actions, two outputs, and two-dimensional private information about the effect of low or high effort on the likelihood of the high output. They provide several insights about optimal menus, including distortion, pooling, and exclusion. Given the difference in environments, ours and their paper are best viewed as complementary. Castro-Pires and Moreira (2021) build on Gottlieb and Moreira (2022) by examining a two-outcome model with moral hazard, adverse selection, and a risk-averse agent. They exhibit conditions under which the optimal compensation scheme is entirely nonresponsive to the type of the agent. Finally, Laffont and Tirole (1986) derive the optimality of linear contracts under moral hazard and adverse selection with a risk-neutral agent, which also decouples in a straightforward way.[5]

We describe the model in Section I. Section II describes decoupling and its implications. Section III derives some results we need in the benchmark cases with only adverse selection or with only moral hazard, while Section IV presents two more necessary conditions of the combined problem. Section V presents *IMC*. Section VI presents the numerical framework that finds the candidate decoupled menu for any given specification of the primitives, and uses it to explore an application. Section VII explores primitives for *IMC*. Section VIII provides extensions, and Section IX concludes. The Appendix contains the proofs of the main results. The online Appendix contains the rest of the proofs as well as the technical details of the numerical analysis.

I. The Model

We analyze a principal-agent problem with both moral hazard and adverse selection. The agent has type $\theta \in [\underline{\theta}, \bar{\theta}]$, with θ distributed according to a cumulative distribution (CDF.) H with strictly positive and continuously differentiable density h. The agent exerts effort $a \in [0, \bar{a}]$, where effort has disutility given by $c(a, \theta)$. The agent is risk averse with strictly increasing, strictly concave, and thrice continuously differentiable utility function u over income.[6] If the agent has type θ, exerts effort a, and receives wage w, then his utility is $u(w) - c(a, \theta)$. He has outside option $\bar{u}$. We assume that u is unbounded below.[7]

Neither a nor θ is observable. Rather, the principal observes a signal x about effort that is distributed according to CDF. $F(\cdot \mid a)$. The principal is risk neutral. Her expected gross benefit from a is equal to $B(a)$, with B concave and strictly increasing. Since neither F nor B depends on θ, our model is one of private values.

[4] Another form of decoupling in a procurement setting quite different from our framework is the incentive-pricing dichotomy in chapter 3 of Laffont and Tirole (1993), which shows when distortions in allocation levels (quantities) can be separated from incentive provision (the cost-reimbursement schedule) in a procurement setting.

[5] An emerging literature on dynamic contracts combines adverse selection and moral hazard. Recent examples are Strulovici (2011); Williams (2015); and Halac, Kartik, and Liu (2016). For tractability, they impose more restrictive assumptions on primitives than we need in our static model.

[6] We use increasing and decreasing in the weak sense, adding "strictly" when needed, and similarly with positive and negative, concave and convex. For any function g, we write $(g)_x$ for the total derivative of g with respect to x, and g_x for the partial derivative. We use $=_s$ to indicate that the objects on either side have strictly the same sign.

[7] In our numerical results we allow for u bounded below but assume that $\bar{u}$ is large enough that the relevant participation constraints bind. This is formally justified in Section VIIB.

The signal x has a positive, twice continuously differentiable density on an interval $[\underline{x}, \bar{x}]$.[8] The density f satisfies the monotone likelihood ratio property (*MLRP*), so that $l(\cdot \mid a) \equiv f_a(\cdot \mid a)/f(\cdot \mid a)$ is increasing in x, where we assume that l is bounded. In some settings, the signal is output, and thus, B is linear in the expectation of x given a. In others, x is a signal distinct from the eventual profits that the principal will realize. The principal's expected profit if the agent exerts effort a and she pays w is $B(a) - w$.

The agent's cost of effort, c, is three times continuously differentiable, with $c(0,\theta) = 0$, and where for all (a,θ) with $a > 0$, we have $c_a > 0$, $c_{aa} > 0$, $c_\theta < 0$, $c_{a\theta} < 0$, $c_{aa\theta} \leq 0$, and $c_{a\theta\theta} \geq 0$. That is, the cost of effort is strictly increasing and strictly convex in effort; total and marginal costs strictly decrease with ability; cost is less convex in effort when ability is higher; and as effort increases, cost becomes more convex in ability.

The contracting problem unfolds as follows. The principal offers a menu of contracts that consists of a pair of functions (π, α), where $\pi(x, \theta_A)$ is the compensation the agent receives if he announces type θ_A and signal x is observed, and $\alpha(\theta_A)$ recommends an effort level given the announced type. Each type of the agent decides whether to accept the menu or take the outside option. If he accepts, he announces a type θ_A and chooses an effort level a. The realization of x is then observed and the agent is paid $\pi(x, \theta_A)$.

As is standard, it is convenient to work with utilities rather than income. Let $v(x, \theta_A) \equiv u(\pi(x, \theta_A))$ be the agent's utility from income when he announces θ_A and the observed signal is x. Let $\varphi \equiv u^{-1}$ be the inverse of u, which is strictly convex since u is strictly concave.[9] Menus are deterministic: there is no randomization over the action and compensation scheme given the type announcement.[10]

By the extended revelation principle (Myerson 1982), it is without loss of generality (*wlog*) to restrict attention to menus of contracts where the agent reports his true type and chooses the recommended effort level. For the bulk of the paper, we assume that the principal wishes all types to participate.[11]

The principal's problem is thus

$$(\mathcal{P}) \qquad \max_{\alpha, v} \int_{\underline{\theta}}^{\bar{\theta}} \left[B(\alpha(\theta)) - \int_{\underline{x}}^{\bar{x}} \varphi(v(x,\theta)) f(x \mid \alpha(\theta)) dx\right] h(\theta) d\theta$$

subject to

$$(IR) \qquad \int_{\underline{x}}^{\bar{x}} v(x,\theta) f(x \mid \alpha(\theta)) dx - c(\alpha(\theta), \theta) \geq \bar{u}, \ \forall \theta,$$

$$(IC) \qquad (\theta, \alpha(\theta)) \in \operatorname*{arg\,max}_{(\theta_A, a)} \int_{\underline{x}}^{\bar{x}} v(x, \theta_A) f(x \mid a) dx - c(a, \theta), \ \forall \theta.$$

[8] At one point in what follows, we consider the case where x has a Bernoulli distribution.

[9] We make the technical assumption that the principal is restricted to measurable functions v such that for each (θ_A, a), $\int v(x, \theta_A) f(x \mid a) dx$ is well defined.

[10] This is often the economically relevant case. In Section VIII, we show that when decoupling works, then the principal is not helped by randomization.

[11] We examine optimal exclusion in Section VIII.

That is, the principal chooses α and v to maximize her expected profit subject to the participation constraint, *IR*, that each type must be willing to accept the menu, and the incentive compatibility constraint, *IC*, that each type is willing to report truthfully and follow the recommended action. A menu (α, v) is *feasible* if it satisfies *IR* and *IC*.

II. Decoupling

The *IC* constraint entails a double continuum of deviations: the agent can *both* misreport their type and then take an action different than the one recommended. This renders problem $\mathcal{P}$ intractable. The heart of this paper is to focus on a *much* simpler problem in which we discard all but two local constraints. In this section we lay out our line of attack.

To begin, return to the moral hazard setting pioneered by Mirrlees (1975) and Holmström (1979). They replace the myriad of incentive constraints—one for each possible effort level—by the first-order condition that the agent does not want to locally change their effort given the compensation scheme they face. Also, as suggested by Grossman and Hart (1983), let us focus first on the cost minimization problem. To do so, let $v^{MH}(\cdot, a, u_0, \theta)$ be the contract that minimizes the cost of implementing a for type θ when the agent is given utility u_0 and the global incentive constraints are replaced by the first-order condition. That is, recalling that $\varphi \equiv u^{-1}$,

$$(\mathcal{P}_{MH}) \qquad v^{MH}(\cdot, a, u_0, \theta) = \arg\min_{z} \int \varphi(z(x)) f(x \mid a) dx$$

subject to

$$\int z(x) f(x \mid a) dx - c(a, \theta) \geq u_0,$$

$$\int z(x) f_a(x \mid a) dx = c_a(a, \theta).$$

Recalling footnote 7, the participation constraint will hold with equality. Let C given by $C(a, u_0, \theta) = \int \varphi(v^{MH}(x, a, u_0, \theta)) f(x \mid a) dx$ be the principal's associated cost function.[12]

We are now ready to describe the relaxed problem that will be central to our analysis. It will pay attention only to two necessary conditions for feasibility. Note first that in $\mathcal{P}$ the agent, having announced his type honestly, should not want to deviate locally in his choice of action. Thus, we must have

$$(IC_{MH}) \qquad \int v(x, \theta) f_a(x \mid \alpha(\theta)) dx - c_a(\alpha(\theta), \theta) = 0, \ \forall \theta,$$

which is the first-order condition in the moral hazard problem for the agent of type θ who is asked to take action $\alpha(\theta)$.

[12] Such a solution is unique if it exists. Mirrlees (1975) and Moroni and Swinkels (2014) point out two existence problems. Boundedness of l rules out the first. We tackle the other in online Appendix Section 7, where we also justify interchanges of differentiation and integration used repeatedly. We henceforth ignore these issues.

To see our second necessary condition, for any given feasible menu (α, v), define

$$S(\theta) = \int v(x,\theta) f(x \mid \alpha(\theta)) dx - c(\alpha(\theta), \theta)$$

as the surplus of the agent of type θ. For (α, v) to be feasible, no type must want to vary his announcement a little while following the recommended action. But, by the envelope theorem, an agent who varies his announcement a little and then takes the recommended action of the announced type changes his payoffs at rate $-c_\theta(\alpha(\theta), \theta)$. Hence, we must have

$$(IC_S) \qquad S(\theta) = \bar{u} - \int_{\underline{\theta}}^{\theta} c_\theta(\alpha(s), s) ds, \ \forall \theta,$$

where surplus is increasing in type since $c_\theta < 0$ and where in any optimum, $S(\underline{\theta}) = \bar{u}$ since otherwise the principal could lower v by a constant while maintaining IC.[13]

Say that the menu (α, v) has the *Holmström-Mirrlees Property* (*HMP*) if for each θ, $v(\cdot, \theta)$ corresponds to the compensation scheme that implements $\alpha(\theta)$ and delivers utility $S(\theta)$ in the relaxed moral hazard problem. That is, $v(\cdot, \theta) = v^{MH}(\cdot, \alpha(\theta), S(\theta), \theta)$ for each θ.

Any solution (α, v) to the problem that maximizes the principal's expected profits subject only to the local conditions IC_{MH} and IC_S above has *HMP*. The idea is that if this is not the case, then we can replace $v(\cdot, \theta)$ by $v^{MH}(\cdot, \alpha(\theta), S(\theta), \theta)$ and save the principal money without violating either condition.[14] Thus, to solve the problem of maximizing the principal's profits subject only to IC_{MH} and IC_S, we can begin by solving the relaxed problem

$$(\mathcal{P}_D) \qquad \max_{\alpha, S} \int_{\underline{\theta}}^{\bar{\theta}} \left[B(\alpha(\theta)) - C(\alpha(\theta), S(\theta), \theta) \right] h(\theta) d\theta$$

subject to

$$IC_S,$$

where the D is mnemonic for "decoupled." This is a standard screening problem, except that S enters the objective nonlinearly. Then for each θ associate the compensation scheme $v(\cdot, \theta) = v^{MH}(\cdot, \alpha(\theta), S(\theta), \theta)$.

This suggests that we attack problem $\mathcal{P}$ with a four-step procedure, *decoupling*:

(i) Solve the moral hazard problem $\mathcal{P}_{MH}$ to derive C for each (a, u_0, θ);

[13] To see this more formally, define $g(\theta_A, \theta) = \int v(x, \theta_A) f(x \mid \alpha(\theta_A)) dx - c(\alpha(\theta_A), \theta)$ as the surplus to type θ of announcing type θ_A and then taking the recommended action for θ_A. Since (α, v) is feasible, $S(\theta) = g(\theta, \theta)$ and $\theta \in \arg\max_{\theta_A \in [\underline{\theta}, \bar{\theta}]} g(\theta_A, \theta)$ for all θ since announcing type θ_A and then taking the recommended action for θ_A is one feasible deviation (of many) for θ. For any given θ_A, $g(\theta_A, \cdot)$ is continuously differentiable with uniformly bounded derivative since c is continuously differentiable on the compact set $[0, \bar{a}] \times [\underline{\theta}, \bar{\theta}]$. Hence, by Theorem 2 of Milgrom and Segal (2002), $S(\theta) = S(\underline{\theta}) - \int_{\underline{\theta}}^{\theta} c_\theta(\alpha(s), s) ds$.

[14] Replacing $v(\cdot, \theta)$ by $v^{MH}(\cdot, \alpha(\theta), S(\theta), \theta)$ at each θ respects IC_{MH} and IC_S since α and S are unaffected and since v^{MH} solves $\mathcal{P}_{MH}$. But, by construction of v^{MH}, this saves the principal money, and strictly so if $v(\cdot, \theta)$ differs from $v^{MH}(\cdot, \alpha(\theta), S(\theta), \theta)$ on a set of types of positive measure.

(ii) Use C as an input into the screening problem $\mathcal{P}_D$;

(iii) Given the solution (α, S) to this problem, for each θ, associate the compensation scheme that solves the relaxed moral hazard problem. That is, $v(\cdot, \theta) = v^{MH}(\cdot, \alpha(\theta), S(\theta), \theta)$; and

(iv) Verify that the solution derived is in fact feasible—and hence, as the solution to a relaxed problem, optimal—in the original problem $\mathcal{P}$. This can be done either by imposing conditions on primitives of the problem or numerically in applications.

A central topic in what follows is to understand Step (iv). We will see that it suffices that α increases sufficiently quickly in θ. This contrasts with a pure screening problem, where α monotone is sufficient (and necessary). We will show that the condition is easily checked in applications and provide the numerical tool kit for doing so. We will also provide economic primitives under which this must be true in any solution to $\mathcal{P}_D$.

The decoupled problem $\mathcal{P}_D$ will be the focus of our analysis henceforth. When decoupling works, problem $\mathcal{P}$, which combines moral hazard with adverse selection, can be solved using a combination of well-understood techniques from each literature separately. There is also significant economic content in knowing when $\mathcal{P}$ decouples. The fact that in many settings the compensation schemes in the optimal menu are Holmström-Mirrlees contracts has implications both theoretical and empirical. If decoupling is valid, then everything we know about each of the two problems separately holds in the combined problem. So, for example, the pattern of distortions in an adverse selection setting comes through, including the optimal downward distortion of actions to reduce information rents. And, everything we know from the moral hazard literature, such as the role of the likelihood ratio in determining the trade-off between incentives and risk bearing, comes through as well. As we will see, there are also interesting interactions between the two forces.

III. The Separate Problems

Because they underlie our analysis of the decoupled problem, Problems $\mathcal{P}_D$ and $\mathcal{P}_{MH}$ are central. In this section, we derive an optimality condition for $\mathcal{P}_D$ that will be used repeatedly below, and we show that a solution to $\mathcal{P}_D$ exists and has convenient differentiability properties. The analysis of $\mathcal{P}_{MH}$ naturally leads to two technical assumptions that will be needed below. Finally, we provide an illustration of how adverse selection and moral hazard can reinforce each other, which tells us that even when Problem $\mathcal{P}$ decouples, there are important interactions between the two frictions. Several of our results for the separate problems are novel to their respective literatures.

A. *Adverse Selection*

Consider first the (relaxed) pure adverse selection problem embedded in $\mathcal{P}_D$, in which the principal chooses effort (allocation) function α and surplus function S.

To uncover the general structure of $\mathcal{P}_D$, consider a general cost function $\hat{C}(a, u_0, \theta)$. When there is moral hazard, $\hat{C} = C$, when effort is observable, $\hat{C} = \varphi(u_0 + c)$, and when the agent is risk neutral, $\hat{C} = u_0 + c$ (whether there is moral hazard or not).[15] Our analysis thus subsumes pure adverse selection with or without quasilinear preferences.

When $\hat{C}$ is linear in u_0, then it is standard to integrate by parts, eliminate the surplus expression, and then maximize the objective function at each type by choice of the action. But, when $\hat{C}$ is not linear in u_0, the cost of asking extra effort from a type depends on his surplus, which by IC_S depends on the effort levels of all lower types. Similarly, the cost of giving extra surplus to higher types depends on their existing surplus. The optimality condition for the relaxed screening problem thus has the somewhat more complicated form

$$(OC) \quad B_a - \hat{C}_a(\alpha(\theta), S(\theta), \theta) + \frac{c_{a\theta}(\alpha(\theta), \theta)}{h(\theta)} \int_\theta^{\bar{\theta}} \hat{C}_{u_0}(\alpha(t), S(t), t) h(t) dt = 0, \ \forall \theta.$$

Note that in the quasilinear case, $\int_\theta^{\bar{\theta}} \hat{C}_{u_0} h = 1 - H(\theta)$, yielding the standard expression. More generally, OC reflects the familiar efficiency versus information rents trade-off. Extra effort by a given type increases match surplus by $B_a - \hat{C}_a$, with h weighting the probability of that type. The extra surplus that must be given to higher types is $c_{a\theta}$. Finally, the cost of providing an extra util to all types above θ is $\int_\theta^{\bar{\theta}} \hat{C}_{u_0} h$.[16,17]

In online Appendix Section 1, we show that a solution (α, S) to IC_S and OC exists, has α continuously differentiable, and solves $\mathcal{P}_D$ as long as $\hat{C}$ is strictly convex in (a, u_0) for each θ and satisfies appropriate boundary conditions in a. We emphasize that this result covers adverse selection with or without moral hazard, and with or without quasilinear preferences, and so substantially expands what is known about existence of an optimal solution in the adverse selection problem. Our method of proof is constructive and central to our numerical analysis in Section VI.

In a pure adverse selection setting, there is also the omitted constraint that α is increasing. When effort is observable, so that $\hat{C} = \varphi(u_0 + c)$, this is the case if h is log-concave and $-c_{a\theta}$ is log-convex in θ (Proposition 3 in the online Appendix). For decoupling to be valid with moral hazard, we will need something stronger than α increasing.

B. *Moral Hazard*

Consider now $\mathcal{P}_{MH}$, the pure moral hazard problem in which the agent's incentive constraint is replaced by the first-order condition. Following Holmström (1979) and

[15] The function $\hat{C}$ could also come from a situation in which the principal is restricted to, for example, linear contracts or simple option contracts.

[16] To see OC formally, rewrite IC_S as $S' = -c_\theta$ for almost all θ and $S(\underline{\theta}) = \bar{u}$, and let $\eta(\theta)$ be the co-state variable associated with $S' = -c_\theta$. Then the Hamiltonian of the problem is $\mathcal{H} = (B - \hat{C})h - \eta c_\theta$, and the optimality conditions are $\partial\mathcal{H}/\partial a = 0$ and $\eta'(\theta) = -\partial\mathcal{H}/\partial S$, along with $\eta(\bar{\theta}) = 0$. Algebra yields OC.

[17] For each θ, equation OC yields a unique solution $\alpha(\theta)$ if $\hat{C}_{aa} > 0$. This is because $\hat{C}_{aa} > 0$ implies that the second term $-\hat{C}_a$ of OC is strictly decreasing in a, and the third term is also decreasing in a since $c_{aa\theta} \leq 0$. Hence, for any given θ, and given $\int_\theta^{\bar{\theta}} \hat{C}_{u_0} h$, if OC yields a solution $\alpha(\theta)$, then this solution is unique. This holds automatically in the pure adverse selection case since $\hat{C}_{aa} = \varphi'' c_a^2 + \varphi' c_{aa} > 0$.

Mirrlees (1975), if λ and μ are the Lagrange multipliers for the participation constraint and the agent's first-order condition, then the cost-minimizing compensation scheme $v^{MH}(\cdot, a, u_0, \theta)$ solves

$$\varphi'\big(v^{MH}(x, a, u_0, \theta)\big) = \lambda + \mu l(x|a), \ \forall x,$$

where λ and μ are functions of a, u_0, and θ and where, since l is increasing in x, so is v^{MH}.

As is well-known, replacing the full set of moral hazard constraints by the first-order condition of the agent need not be valid. Say that the *First-Order Property* (*FOP*) is satisfied if the agent's expected utility is concave in his effort given v^{MH}.[18] There are well-known primitive conditions for *FOP* to hold.[19]

ASSUMPTION 1 (FOP): *The First-Order Property holds.*

Given the analysis of $\mathcal{P}_D$, we will also need that the cost function C is convex in the action.[20]

ASSUMPTION 2 (Convexity of C): $C_{aa} > 0$.

Primitives for this intuitive property are not trivial to find. One set is provided by Jewitt, Kadan, and Swinkels (2008), who focus on the local informativeness of the signal. Their condition is satisfied in the linear probability case where $F_{aa} = 0$ (Jewitt, Kadan, and Swinkels 2008, Example 2). Chade and Swinkels (2020a)—henceforth, *CS*—show that under mild conditions on the utility function, Assumption 2 holds if $\bar{u}$ is sufficiently large.

C. *Interacting Distortions*

Without adverse selection, the principal chooses effort for each type to set $B_a\big(\alpha(\theta)\big) - \hat{C}_a\big(\alpha(\theta), \bar{u}, \theta\big) = 0$. The expression in *OC* differs from this in two ways. First, there is the term $-\left(c_{a\theta}/h\right)\int \hat{C}_{u_0} h$. As is standard in screening problems, this term disappears for the highest type but distorts effort down for lower types so as to lower the information rents of higher types. But here, there is the second difference that rather than being evaluated at utility $\bar{u}$, $\hat{C}_a$ is evaluated at $S(\theta)$. In the full-information case, higher information rents raise the marginal cost of inducing effort since $\hat{C} = \varphi(u_0 + c)$, and so $\hat{C}_{au_0} = \varphi'' c_a > 0$. Under reasonable conditions, $\hat{C}_{au_0} > 0$ also holds when there is moral hazard (that is, when $\hat{C} = C$).[21] There is then an additional force pushing effort down from the full-information case even for the highest type, and the distortions that arise via the effect of information rents on $\hat{C}_a$ *reinforce* the distortions from adverse selection.

[18] That is, $\int v^{MH}(x, a, u_0, \theta) f(x|\cdot) dx$ is concave for each (a, u_0, θ).

[19] The simplest condition for this is that $F_{aa} \geq 0$, the convexity of the distribution function condition (Rogerson 1985). See also Jewitt (1988), and a large literature that follows, and Chade and Swinkels (2020b) for a summary.

[20] The other conditions for $C(\cdot, \cdot, \theta)$ to be convex will play a limited role, and we will introduce them as needed.

[21] Primitives are provided in online Appendix Section 2, Lemma 10 and by *CS* when $\bar{u}$ is large.

IV. Two More Necessary Conditions

There are two more necessary conditions inherent in the agent's problem in $\mathcal{P}$: the first-order condition with respect to misreporting the agent's type while holding fixed his action and a second-order condition. Each will be used extensively in what follows, where the latter will allow us to derive another implication of the combined presence of adverse selection and moral hazard and will be used to show that our central sufficient condition for feasibility is also necessary in the important case that the distribution over the signal moves linearly in effort.

To derive our first condition, note that the agent must not benefit locally from misreporting his type while holding fixed his action. Since the agent's utility from announcement θ_A and action $\hat{a}$ is $\int v(x,\theta_A)f(x\,|\,\hat{a})dx - c(\hat{a},\theta)$, this tells us that

$$(IC_A) \qquad \int v_\theta(x,\theta)f(x\,|\,\alpha(\theta))dx = 0$$

for each θ where v is differentiable.[22] To understand the relationship between this first-order condition and the two previously derived, note that under differentiability of α and v, any two of IC_{MH}, IC_S, and IC_A imply the third. To see this, differentiate $S(\theta) = \int v(x,\theta)f(x\,|\,\alpha(\theta)) - c(\alpha(\theta),\theta)$ to arrive at

$$0 = -[S'(\theta) + c_\theta] + \int v_\theta f + \alpha'\Big(\int v f_a - c_a\Big),$$

where the first term being zero is equivalent to IC_S, the second to IC_A, and the third to IC_{MH}. The geometric intuition is that each deviation corresponds to a different direction away from $(\theta,\alpha(\theta))$ in the two-dimensional space of announcements and actions.

There is also a useful second-order condition. If v is twice differentiable in θ, then, using IC_{MH} and IC_A, Lemma 1 in Appendix A shows that the second-order conditions for the local concavity of the agent's surplus at $\theta_A = \theta$ and $\hat{a} = \alpha(\theta)$ reduce to the single condition

$$(SOC) \qquad \int v_\theta(x,\theta)f_a(x\,|\,\alpha(\theta))dx \geq 0.$$

This says that if type θ raises his announcement by a little, then the compensation scheme he faces changes so as to strengthen his marginal incentive for effort.

Another Interaction.—The intuition for *SOC* is that if this condition is not met, then a double deviation of announcing a lower than truthful type and then taking a higher than recommended action is attractive. Indeed, an implication of *SOC* is that the effort schedule must be strictly increasing. This stands in contrast to the pure screening case, where the effort schedule must be increasing, but not strictly so, illustrating that the implications of the joint problem are more than just those of each problem separately. To see the underlying economics, assume that two types take the

[22] Online Appendix Section 7 shows that λ and μ, and thus $\tilde{v}$ and C, are twice continuously differentiable in their arguments on the interior of the set where they are defined and that therefore from OC, any solution to $\mathcal{P}_D$ that lies in this set will have α and v continuously differentiable.

same action. Holding the action fixed at the recommendation, each type is of course indifferent to announcing the other's type. But, since the higher type has a strictly lower marginal cost of effort, it must be that the contract facing the lower type provides strictly stronger incentives for effort than that facing the higher type. But then, the high type, who by construction is indifferent about his effort choice given the low-powered incentive scheme, can benefit by announcing the low type and then taking advantage of the more powerful incentives by deviating to a higher action.

Indeed, from this logic, one can derive a simple lower bound for how fast effort must increase in any incentive-compatible menu. To see this, differentiate IC_{MH}, which states that $\int v f_a = c_a$ for all θ, to arrive at $\int v_\theta f_a = \alpha' c_{aa} + c_{a\theta} - \alpha' \int v f_{aa}$. But then, *SOC* holds if and only if

$$\alpha' \geq \frac{-c_{a\theta}}{c_{aa} - \int v f_{aa}} > 0, \tag{1}$$

where the denominator is positive by the agent's second-order necessary condition for his optimal choice of action and the numerator is strictly positive. It is worth noting that while actions must strictly increase in type, this does not imply that different types receive different compensation schemes. A range of types can receive the same compensation scheme, with (1) holding as an equality for that range of types.

V. A First Sufficient Condition: Increasing Marginal Costs

We now provide our first sufficient condition for decoupling.

DEFINITION 1 (*IMC*): *Action schedule α satisfies the increasing marginal cost condition (IMC) if $c_a(\alpha(\cdot), \cdot)$ is increasing, so that more capable agents face stronger incentives for effort.*

That is, the action increases fast enough in θ that the increase in marginal disutility from the increase in the action offsets the decrease in marginal disutility from the higher ability of the agent. Indeed, where α is differentiable, what is required is that $\alpha' \geq -c_{a\theta}/c_{aa}$.[23] Note that under *FOP*, $-\int v f_{aa} \geq 0$, and so our condition is stronger than the necessary condition derived in (1). Of course, if $f_{aa} = 0$, then the two conditions coincide. We return to this important case in Sections VII and VIII.

Recall that Assumption 1 states that *FOP* holds. We now show that satisfying *IMC* in $\mathcal{P}_D$ is sufficient for decoupling to be valid.

THEOREM 1 (Sufficiency of *IMC*): *Let FOP hold, let α be continuous and satisfy IMC, let S satisfy IC_S, and let $v(\cdot, \theta) = v^{MH}(\cdot, \alpha(\theta), S(\theta), \theta)$ for each θ. Then, (α, v) is feasible in $\mathcal{P}$. Thus, if (α, S) is optimal in $\mathcal{P}_D$, it is optimal in $\mathcal{P}$.*

The role of *IMC* is to show that double deviations by the agent facing (α, v) are suboptimal. To see the idea, see Figure 1, which plots types on the horizontal axis and actions on the vertical axis and shows the graph of α. For any given true type θ_T

[23] *IMC* automatically holds at jumps in α, and since α is monotone, it is almost everywhere differentiable.

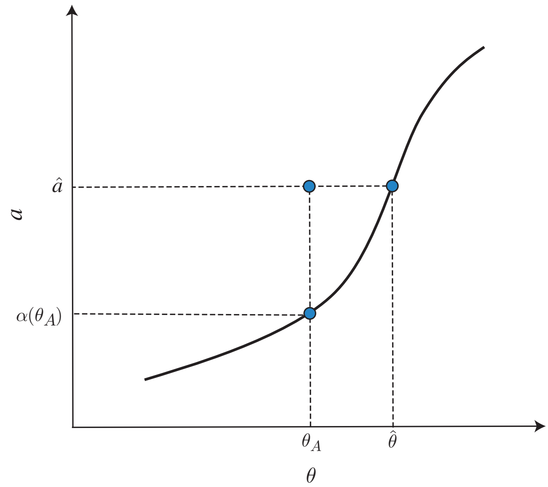

Figure 1. *IMC*

Note: Under *IMC*, a deviation to $(\theta_A, \hat{a})$ is dominated by the on-graph deviation $(\hat{\theta}, \hat{a})$, which in turn is dominated by telling the truth and taking the recommended action.

of the agent, every point where the announced type is between $\underline{\theta}$ and $\overline{\theta}$ and the action is between zero and $\bar{a}$ is a feasible deviation. The agent can lie about their type and then take any action they desire. We need to rule out all such deviations. The value of θ_T will not be germane in what follows and so is not shown in the figure.

Note first that by construction of S, and recalling IC_A, the agent does not want to locally change his type announcement given that he intends to take the recommended action. That is, the agent does not want to locally move along the graph of α. But then a standard argument (Lemma 2 in Appendix A) shows that conditional on taking the recommended action—i.e., conditional on being on the graph of α—the agent is best-off to announce his type honestly. The idea is that moving along the graph of α by increasing the announcement beyond θ_T involves extra effort that for each θ, type θ is indifferent to, and so one that θ_T strictly dislikes, and similarly for announcements below θ_T.

The heart of the proof is to show that for any "double deviation" off the graph that the agent might contemplate, there is an even better deviation *on* the graph. Since the agent's favorite point on the graph of α is to tell the truth, this will establish that he does not want to deviate at all.

To see the argument, imagine that type θ_T is contemplating the double deviation of announcing θ_A and then taking action $\hat{a}$ above $\alpha(\theta_A)$ (the case $\hat{a}$ below $\alpha(\theta_A)$ is similar). We will show that regardless of θ_T, compared to $(\theta_A, \hat{a})$, an even better deviation is $(\hat{\theta}, \hat{a})$. That is, given that they intend to take action $\hat{a}$, the agent should increase their announcement of type so as to get back on the graph of α.

To see that increasing the announcement to $\hat{\theta}$ benefits the agent, let us consider moving from $(\theta_A, \hat{a})$ to $(\hat{\theta}, \hat{a})$ by first moving vertically down to the graph of α, that is, to the point $(\theta_A, \alpha(\theta_A))$, and then moving back up to effort $\hat{a}$ by traveling up and to the right along the graph of α to $(\hat{\theta}, \hat{a})$. Since we start and end at the same effort level $\hat{a}$, it follows that the agent of type θ_T incurs the same cost $c(\hat{a}, \theta_T)$ at both points. Hence, all that matters is to show that the expected utility of income, $\int vf$, rises on net as one travels this path.

To do this, let $\tilde{c}_a = c_a(\alpha(\theta_A), \theta_A)$ be the marginal disutility of effort to type θ_A when they take their recommended action. We will show that as one drops vertically, $\int vf$ is falling at rate *at most* $\tilde{c}_a$, while as one moves back up along the graph, $\int vf$ is increasing at rate *at least* $\tilde{c}_a$. Thus, the total fall in $\int vf$ from the vertical move is offset by the total increase in $\int vf$ along the graph of α when moving toward $(\hat{\theta}, \hat{a})$.

Consider first the vertical drop. By the first-order condition on effort, *IC_{MH}*, $\int vf_a$ is equal to $\tilde{c}_a$ on the graph of α at $(\theta_A, \alpha(\theta_A))$. But, by *FOP*, the expected utility from income for any given announcement is concave in the action, and so $\int vf_a$ is lower at higher efforts, and so as one drops vertically, $\int vf$ is falling at rate at most $\tilde{c}_a$, as claimed.

Consider now moving back up and to the right along the graph of α. By *IC_A*, $\int v_\theta f = 0$, and thus, the horizontal movement at each point along the graph has zero marginal impact on the expected utility of income. Hence, the change in $\int vf$ from moving along the graph of α is simply equal to the impact of a vertical movement at that point, given by $\int vf_a$. But, by *IC_{MH}*, this impact is equal to the marginal cost $c_a(\alpha(\theta), \theta)$ of the type $\theta > \theta_A$ active at that point. By *IMC*, since the marginal cost of the active type increases along the graph, this is bigger than $\tilde{c}_a$, and we are done. Thus, for any double deviation, there is an even better on-graph deviation available.

In Theorem 4 of the online Appendix, we extend this result to allow for jumps in α. These will occur if the principal faces a menu cost or is otherwise limited to a finite number of compensation schemes, or if C is ill behaved.

VI. Numerical Decoupling

Decoupling is not just a theoretical tool. In this section, we show that each component of the decoupling program is amenable to numerical analysis. In particular, our proof of existence of a solution to $\mathcal{P}_D$ points the way to numerically efficient calculation of a solution to the relaxed adverse selection problem given C, where C itself is tractable because it is the solution to a relaxed problem. Based on this, we describe and implement an efficient algorithm to implement decoupling and numerically calculate a candidate menu for any given specification of the primitives of the model. It is then straightforward to verify whether the candidate menu satisfies *IMC*. In other words, Theorem 1 can play an important role as a *verification theorem* for applied research.

In this section, we first describe the algorithm and its numerical implementation. Using these numerical tools, we solve and analyze the properties of the decoupled problem in an applied example.

A. *Algorithm and Numerical Implementation*

Here, we sketch the algorithm that generates a candidate menu. See online Appendix Section 4 for details, and see the Python code we provide that operationalizes the algorithm. To begin, let $\rho = (\varphi')^{-1}$ be the function that takes in a value for $1/u'$ and returns a value for u. Then, for each triple (a, u_0, θ), one can solve the relaxed moral hazard problem by finding two Lagrange multipliers (λ, μ) that satisfy

$$\int \rho(\lambda + \mu l(x|a)) f(x|a) dx = c(a, \theta) + u_0 \tag{2}$$

and

$$\int \rho(\lambda + \mu l(x|a)) f_a(x|a) dx = c_a(a, \theta). \tag{3}$$

One can then derive simple expressions for C, C_a, and C_{u_0} in terms of λ and μ.[24]

Now, consider attempting to solve for α using these equations and OC. This is made complicated by the fact that OC depends on the actions of types both above and below θ. To cut this knot, let $Z(\theta) \equiv \int_\theta^{\bar{\theta}} C_{u_0}(\alpha(t), S(t), t) h(t) dt$ so that OC can be rewritten as

$$B_a(\alpha(\theta)) - C_a(\alpha(\theta), S(\theta), \theta) + \frac{c_{a\theta}(\alpha(\theta), \theta)}{h(\theta)} Z(\theta) = 0, \tag{4}$$

and note that for any given values of S and Z, and under the assumptions of our existence result, α, λ, and μ are tied down by this equation along with (2) and (3) in a way that is numerically easy to compute. We thus have a pair of differential equations, one for S and one for Z, where S has equation of motion $S' = -c_\theta$ with boundary condition $S(\underline{\theta}) = \bar{u}$ and Z has equation of motion $Z' = -C_{u_0} h$ with boundary condition $Z(\bar{\theta}) = 0$. Indeed, the heart of our proof of existence of a solution to $\mathcal{P}_D$ is to establish conditions under which this pair of differential equations has a solution.

To solve this pair of differential equations, we discretize the type space, set $Z(\bar{\theta}) = 0$, and guess a value of the surplus $S(\bar{\theta})$ for the highest type. Then, we solve iteratively from top to bottom to find the surplus, the effort, and the multipliers for all lower types. As in the proof of existence of a solution to $\mathcal{P}_D$, for a large enough guess of $S(\bar{\theta})$, $S(\underline{\theta})$ will overshoot $\bar{u}$, and for a small enough guess, it will undershoot. We then numerically find the $S(\bar{\theta})$ such that $S(\underline{\theta}) = \bar{u}$, at which point we have a candidate solution. The final step is checking whether the candidate solution satisfies *IMC*. If it does, then we have found the solution to the original problem.

B. *Computational Advantages*

When it works, decoupling has major computational advantages compared to brute-force searching for the optimal menu in a discretized setting. The characterization of the compensation scheme for a given type, surplus, and effort reduces to finding two numbers, the Lagrange multipliers, instead of an entire mapping from output to payments. Solving the system of differential equations is also fast in practice because we generate our candidate menu by imposing only two local conditions.

[24] Recall that our approach requires that the *FOP* holds. In our numerical implementation, we assume a distribution that belongs to a class that assures the validity of the FOP (see LiCalzi and Spaeter 2003). Whenever the primitives of the model do not guarantee the validity of the *FOP*, one must check its validity numerically.

Consider instead a brute-force approach to finding an optimal menu when there are n types, effort levels, signals, and wage levels. There are n^n mappings from signals to wages, and thus, since the principal may wish to choose a distinct compensation scheme for each announced type, at least n^n choose n different collections of incentive schemes that the principal might choose. When n is 5, this equals 10^{15}, and when n is 10, this is 10^{93}. For each such collection of compensation schemes, we need to calculate the optimal announcement and action chosen by each type of the agent so as to calculate the profit of the principal.

It is also unclear what such an exercise would teach us, even if it was feasible. The results of a brute-force approach do not reveal anything about the underlying structure of the optimal menu or how one might expect that menu to vary with underlying primitives. Our algorithm is flexible enough to allow the analyst to input their preferred primitive functions and parameters and solve for the optimal mechanism and fast enough to explore alternative specifications efficiently. Of course, our algorithm is silent if *IMC* (or the alternative condition *SCC* discussed in Section VIII) fails to hold for the numerically calculated candidate menu. But, as we next illustrate, our method allows us to solve and analyze problems with simultaneous moral hazard and adverse selection that would otherwise be computationally infeasible.

C. *Numerical Application*

Consider the paradigmatic problem of a firm designing incentives for a prospective CEO (or of a venture capitalist designing incentives for an entrepreneur). An enormous literature has studied this problem from a moral hazard point of view, where the key question is how to structure incentives to motivate effort (see Frydman and Jenter 2010; Murphy 2013; and Edmans and Gabaix 2016 for surveys). The problem has also been studied from a screening point of view, where one question is how to tailor effort to ability, and a key question is often how to motivate a CEO of low talent not to apply or an entrepreneur with a poor project not to pursue funding (see Manove, Padilla, and Pagano 2001; Kaplan and Stromberg 2001; and Armstrong, Hail, and Zhang 2022). But real-world incentive schemes have to solve both problems at once. Apart from simple two-type by two-action by two-outcome examples, the problem in which both frictions are present at once is unexplored.

We now illustrate how to numerically analyze this problem. The primitives that describe this interaction are the distribution of performance given effort F, which we take as $F(x \mid a) = xe^{(x-1)(1+a)}$, where $x \in [0,1]$ and $a \in [0,1]$; the firm's benefit function from effort B, which we take as $B(a) = b_0 + 50000 \cdot E[x \mid a]$; the CEO's utility function u over income and outside option $\bar{u}$, where we take $u(w) = 4w^{1/5}$ and $\bar{u} = 25$; her effort cost c, which we take as $c(a,\theta) = (2-\theta)ae^{(a^2-1)}$, where $\theta \in [0,1]$; and the type distribution h, which we take as $h(\theta) = 1/3 + 2\theta^2$.[25] Our machinery allows the modeler to vary these objects to suit the context at hand. For this section we set $b_0 = 0$. We vary b_0 in Section VIII when we discuss optimal exclusion.

[25] The output distribution F was selected from a class proposed by LiCalzi and Spaeter (2003) that ensures *FOP*.

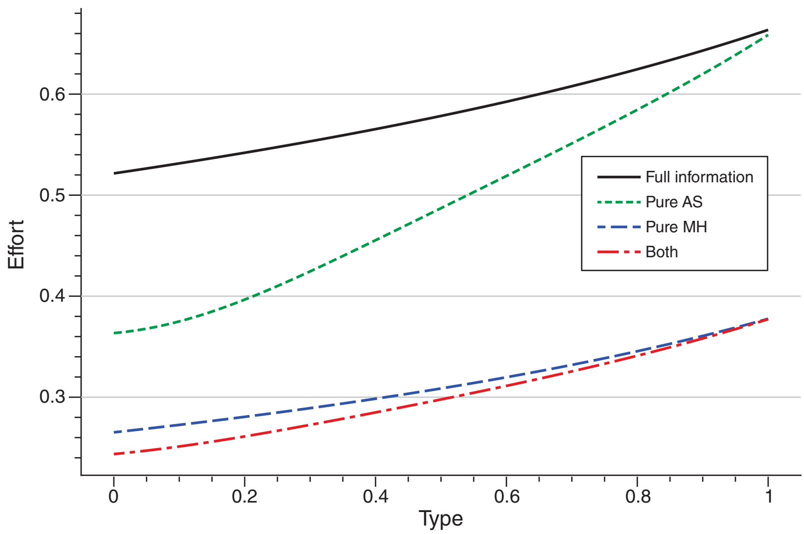

Figure 2. Effort Levels

We apply decoupling to this parameterization using our numerical algorithm and show that the decoupled solution satisfies *IMC* and so is valid. For comparison, we also compute the full-information case where both effort and ability are observable, the pure moral hazard case, and the pure adverse selection case.

Figure 2 depicts the effort level for each type in each of the cases. The solid black line denotes the full-information case, the dotted green line the pure adverse selection case, the dashed blue line the pure moral hazard case, and the dash-dotted red line the case in which moral hazard and adverse selection are simultaneously present.

The figure reveals several intuitive properties. Compared to full information, under pure adverse selection, the firm distorts the effort of all types of the CEO downward to reduce the information rents of higher types. This reflects the usual effect from the efficiency versus information rents trade off. Moreover, the allocation of the highest type is efficient in the sense that it equates the firm's marginal benefit of effort with the marginal cost of inducing it. When the agent is risk neutral, this implies the same effort level as in the full-information case. But, as discussed above, since the CEO is risk averse, the marginal cost of inducing more effort is higher when she is better-off. As the highest type receives strictly positive information rents, there is thus an additional force pushing down the effort of the most capable type. This explains why the green dotted line lies below the solid black line even at the highest type.

Comparing pure moral hazard with full information, one observes the usual risk versus incentives trade-off. To provide incentives, the firm conditions CEO pay on output, which introduces risk and increases the marginal cost of inducing effort. Hence, the effort implemented in the pure moral hazard case for each CEO type is everywhere below the effort in the full-information benchmark.

Finally, in this example, when both frictions are present, the distortions reinforce each other, and effort is everywhere below either of the cases with a single

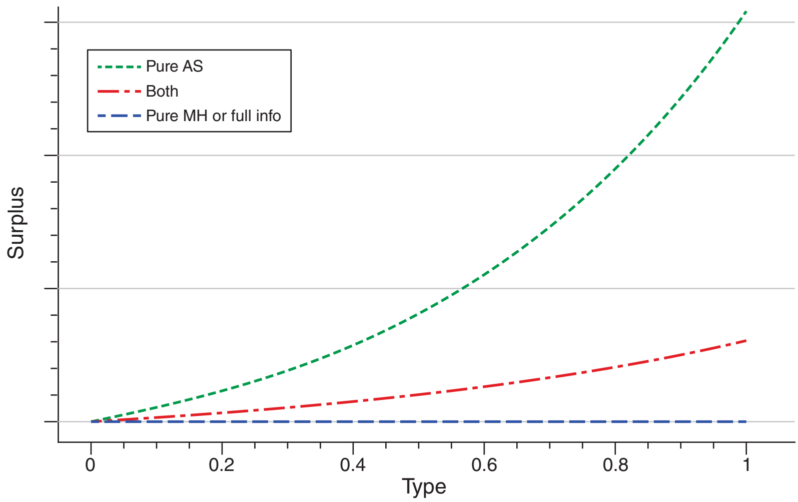

FIGURE 3. SURPLUS LEVELS

distortion. Consistent with Section IIIC, compared to the pure moral hazard case, a further downward distortion in effort reduces the information rents left to higher types.[26] Moreover, since $S_\theta = -c_\theta$, an implication of the fact that under the combined frictions the CEO exerts less effort than under pure adverse selection is that the surplus of the CEO rises less quickly in her type under the combined frictions than under pure adverse selection. So, while compared to full information (or pure moral hazard) the CEO benefits from uncertainty about her type, she is harmed if in addition there is uncertainty about her effort. Figure 3 illustrates the surplus of the agent as a function of her type when her type is known (the horizontal dashed blue line at $\bar{u}$), when there is pure adverse selection (the dotted green line), and when there is both adverse selection and moral hazard (the dash-dotted red line).

Figure 4 displays the compensation scheme for each of several types in each of the three cases with an informational friction. In the pure adverse selection case, each compensation scheme consists of a fixed wage, with higher types of the CEO getting a higher wage to compensate her for her increased effort and to reflect her increased information rents. In the other two cases, compensation depends on realized output. But when both frictions are present, compensation is less steep in output than when only moral hazard is present, reflecting the further downward distortion in effort in this case.

Recall that, under decoupling, the compensation schemes for each type are of the Holmström-Mirrlees form. Thus, for each type, the CEO's compensation is pinned down by likelihood ratios and the Lagrange multipliers. Figure 5 depicts the multipliers associated with pure moral hazard and those associated with the combined problem. The multiplier μ on the incentive constraint is everywhere lower with the combined frictions. This is intuitive, as less effort is being asked of the CEO. Interestingly, the multiplier λ on the participation constraint is higher with both

[26] It is less obvious that effort with both moral hazard and adverse selection is below that with pure adverse selection since as is well-known, the marginal cost of inducing effort may be lower with moral hazard than without.

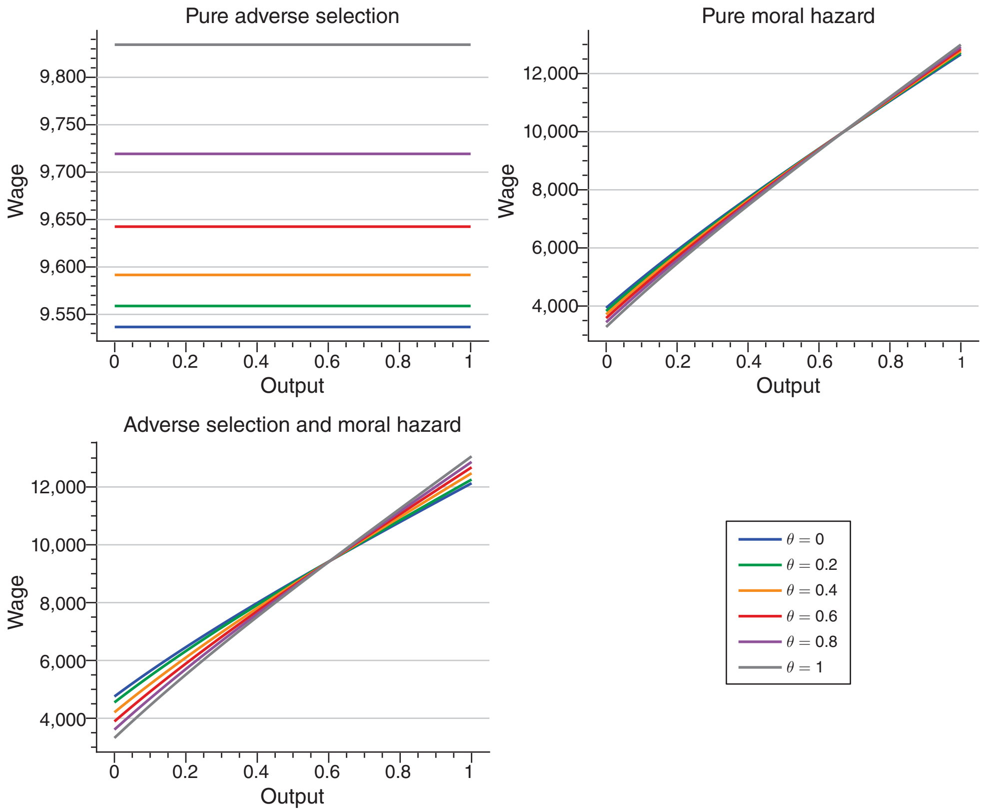

Figure 4. Compensation Schemes

frictions for high types of the agent but lower for low types of the agent. The conflicting forces are that the CEO has higher utility with both frictions than with pure moral hazard, and so, all else equal, the participation constraint is harder to satisfy. But, with both frictions, the CEO is asked to exert less effort and so faces less risk, which makes the participation constraint easier to satisfy.

It remains to check that *IMC* holds for the candidate menu given by decoupling. This is demonstrated by Figure 6, which shows that $c_a\big(\alpha(\cdot), \cdot\big)$ is increasing. Hence, the decoupled solution is optimal in the original problem.

Although we have presented the three information cases separately, decoupling also allows us to smoothly parameterize them. As detailed in online Appendix Section 4.8, one can parameterize the problem in such a way that when $\tau = 0$, the distribution of output becomes degenerate at the effort of the agent, while when $\tau = 1$, the disutility of effort becomes independent of θ. Hence, $\tau = 0$ is the pure adverse selection benchmark, $\tau = 1$ is the pure moral hazard benchmark, while when $\tau \in \big(0, 1\big)$, both frictions are at play.

VII. Primitives for *IMC*

In the previous section, we showed how to use Theorem 1 as a verification tool. Here, we present two sets of conditions on the primitives of the model under which

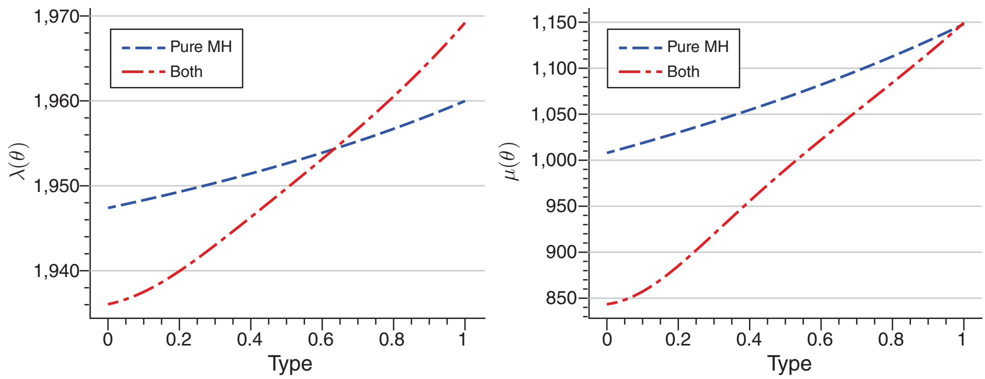

FIGURE 5. MULTIPLIERS

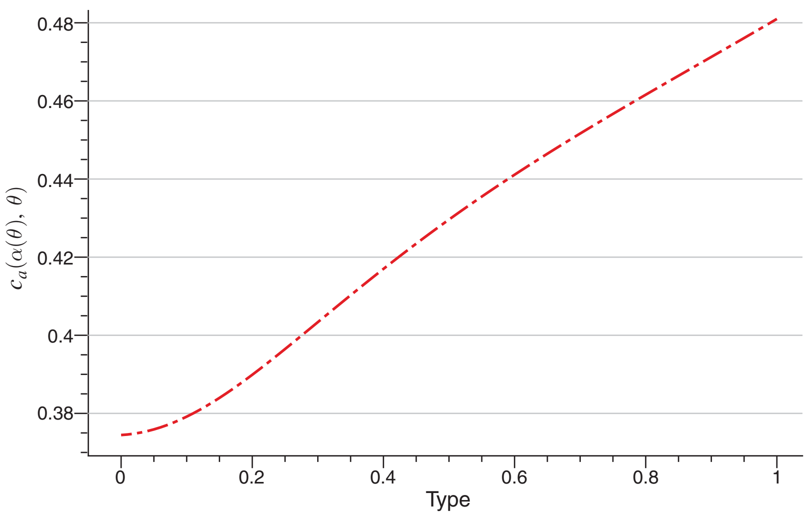

FIGURE 6. *IMC*

the solution to $\mathcal{P}_D$ is guaranteed to satisfy *IMC,* and so the associated menu will solve $\mathcal{P}$.

A. *Linear Output*

We begin with the *linear probability* case in which the probability of an outcome below any given threshold is linear in effort, so that $F_{aa} = 0$. This is equivalent to F being a linear combination of two distributions (spanning). One common application is when output is binary, as for example in an insurance setting where an accident either does or does not occur. A second is when the signal is continuously distributed with density given by $f(x|a) = (1-a)f_L(x) + af_H(x)$ for each x and a and where f_H/f_L is increasing to ensure *MLRP*.[27] Note that *FOP* is automatic in the

[27] Spanning information structures were first used in moral hazard models by Grossman and Hart (1983) and significantly further analyzed by Kirkegaard (2017), who studies moral hazard under a more general spanning

linear probability setting since the agent's expected utility from income is linear in his effort.

To exhibit primitives under which *IMC* holds at a solution to $\mathcal{P}_D$, it is convenient to assume that B is linear in a. This is automatic in the canonical case where $B(a) = E[x \mid a]$.

ASSUMPTION 3: *B is linear in a.*

Our next two assumptions control two sets of forces.

ASSUMPTION 4: *For each* a, $-c_{aa}h/c_{a\theta}$ *is increasing in* θ.

This holds if, for example, $c(a,\theta) = (1-\theta)a + a^2$ and h is increasing.

ASSUMPTION 5: *For each* θ, $c_{aa}/c_{a\theta}$ *is increasing in a.*

Recall from Section VIA that ρ maps $1/u'$ into utility. Our next result assumes that ρ is concave, which holds for most commonly used utility functions.

THEOREM 2 (Linear Output): *Let Assumptions 2–5 hold, let* $F_{aa} = 0$, *let* $\text{skew}_F(l) \le 0$, *and let* ρ *be concave. Then, any solution to* $\mathcal{P}_D$ *satisfies IMC. Hence, its associated menu is optimal in* $\mathcal{P}$.

The proof is in Appendix A.A3 and depends on a result in *CS* and a covariance inequality. As we show in Lemma 11 in the online Appendix, negative skewness of l holds if $f(\cdot \mid a) = af_H(\cdot) + (1-a)f_L(\cdot)$, where $\text{skew}_{F_L}(x) \le 0$ and f_H/f_L is increasing and concave.[28]

B. *A Large Outside Option*

Our second set of primitives for *IMC* apply when the outside option of the agent is large enough. Of course, as the outside option grows, the expense of employing the agent rises, and so this is only relevant if the value to the principal of hiring the agent is also large. A realistic example is a large firm hiring a CEO with attractive outside opportunities.

We will require some mild conditions on u as the utility of the agent grows. Recall that the coefficient of absolute prudence, P, is $-u'''/u''$, and of absolute risk aversion, A, is $-u''/u'$.

ASSUMPTION 6: *As* $w \to \infty$, *we have that* $u \to \infty$, $u' \to 0$, $A/u' \to 0$, *and* P/A *has a finite positive limit.*

---

condition under which the first-order approach may not be valid. Gottlieb and Moreira (2022) use the spanning condition in their principal-agent problem with adverse selection and moral hazard but, unlike our setting, with a risk-neutral agent subject to limited liability, and show that offering a single compensation scheme can be optimal.

[28] In the two-outcome case, *IMC* and $C_{aa} > 0$ hold under Assumption 4, and $(3a-2)/[a(1-a)] \le (c_{aaa}/c_{aa}) \le (c_{aa\theta}/c_{a\theta}) + (3a-1)/[a(1-a)]$ for all a, which is satisfied if $c(a,\theta) = (2-\theta)a^2$, $\theta \in [0,1]$, and $a \in [1/3, 2/3]$. See online Appendix Section 5 for details.

The most central of these assumptions is that $A/u' \to 0$. Since $A/u' = (1/u')'$, the agent becomes closer to risk neutral over intervals of wealth of any fixed length as wealth grows.[29]

Per Assumption 3, let $B(a) = \beta_0 + \beta_1 a$. We will make the following assumption.

ASSUMPTION 7: $c_a(0,\cdot) = 0$ *and* $0 < \beta_1 < \varphi'(\bar{u})c_a(\bar{a},\bar{\theta})$.

For each θ, let $\tilde{\alpha}(\theta)$ solve

$$(OC_{lim}) \qquad \frac{\beta_1}{\varphi'(\bar{u})} - c_a(\tilde{\alpha}(\theta),\theta) + c_{a\theta}(\tilde{\alpha}(\theta),\theta)\frac{1 - H(\theta)}{h(\theta)} = 0,$$

where Assumption 7 ensures that the solution is interior. We then have the following result.

THEOREM 3 (Large Outside Option): *Let Assumptions 1–4, 6, and 7 hold, and let F be $\mathcal{C}^4$. Then for all $\bar{u}$ sufficiently large, the solution to $\mathcal{P}_D$ satisfies IMC, and so its associated menu is optimal in $\mathcal{P}$. The action schedule converges uniformly to $\tilde{\alpha}$ as $\bar{u}$ diverges.*

The proof is in Appendix A.A3, which uses a weaker version of Assumption 4. Note that OC_{lim} reflects a pure screening problem with quasilinear utility. Crucially, it can be solved *pointwise* for $\tilde{\alpha}$, with the complexities of S and $\int C_{u_0} h$ falling away.

For some intuition, under Assumption 6, as the outside option increases, the agent becomes less risk averse. This mitigates the moral hazard problem, and as a result, the problem becomes closer to a pure adverse selection problem. But, since moral hazard disappears only in the limit, we still need to ensure that the full set of constraints is satisfied, and this is guaranteed by *IMC*, which holds for sufficiently large $\bar{u}$ under the premises of Theorem 3.

A concern left open by Theorem 3 is that *IMC* holds only for extremely high values of $\bar{u}$, when the agent is close to risk neutral. Our numerical example shows that this is not the case. Define $R(\theta) = \varphi'(v(\bar{x},\theta))/\varphi'(v(\underline{x},\theta))$ as the ratio of marginal utilities between the lowest and highest payment for type θ for the compensation scheme v obtained from decoupling. Under risk neutrality, R is one. Figure 7, however, shows that R ranges from 2.1 for the lowest type to 2.97 for the highest type for a case where $\bar{u}$ is large enough for *IMC* to hold, and thus, the agent is far from risk neutral.

Theorem 3 and OC_{lim} allow for comparative statics analysis of stochastic shifts in the distribution of types. Proposition 1 in Appendix A.A3 establishes that when the distribution of types shifts in a precise sense toward higher types, then distortions grow. While intuitive, they are impossible to establish without first knowing that decoupling holds when $\bar{u}$ is sufficiently large, which then allows us to use OC_{lim} to pin down the limiting behavior of the solution.

[29] Assumption 6 is satisfied for the *HARA* utility functions $u = (1-\gamma)\left[aw/(1-\gamma) + b\right]^{\gamma}/\gamma$ with parameter $\gamma \in (0,1)$ but fails for $u = \log w$.

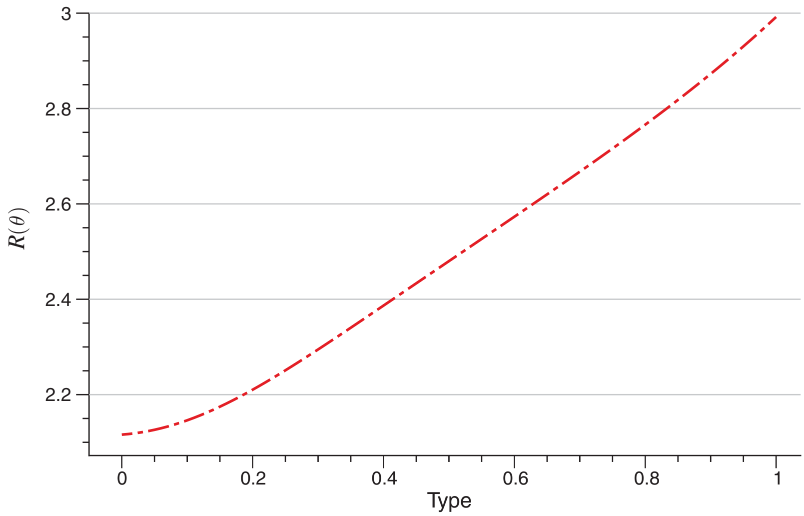

Figure 7. Risk Aversion Measure

We stress that the premises of Theorem 3 hold for standard classes of utility and cost of effort functions, coupled with the standard conditions on the distribution of types. Taken together with our numerical results, Theorems 2 and 3 suggest that decoupling holds quite broadly.[30]

VIII. Extensions

We now introduce six extensions to our analysis. The first builds on the fact that *IMC* is necessary for incentive compatibility in the linear output case, and derives an optimality condition that takes into account *IMC* as an explicit constraint in problem $\mathcal{P}_D$. The second provides an alternative sufficient condition for feasibility. The third explores common values. The fourth examines optimal exclusion in the general setting. The fifth shows that random mechanisms are superfluous when decoupling is valid. Finally, we examine the extension of our results to a social planner. We outline the results here. See online Appendix Section 6 for details.

Linear Output Case and Necessity of IMC.—Recall that in the linear case, the necessary condition *SOC* is equivalent to *IMC*. Thus, (α, v) is feasible in $\mathcal{P}$ *if and only if* it satisfies IC_S, IC_{MH}, and *IMC*. But then, any optimal solution to $\mathcal{P}$ has the property that compensation schemes are of the Holmström-Mirrlees form. Thus, solving $\mathcal{P}$ is equivalent to solving

$$(P_L) \qquad \max_{\alpha,S} \int (B - C)h$$

subject to

$$IMC \quad \text{and} \quad IC_S,$$

[30] In our example, *IMC* fails if the outside option is sufficiently low. So neither can *IMC* be taken for granted.

and then for each θ, associating the compensation scheme $v^{MH}(\cdot, \alpha(\theta), S(\theta), \theta)$.

Even if *IMC* binds over some intervals, P_L is much more tractable than $\mathcal{P}$. It is a fairly standard screening problem, simply with monotonicity of α replaced by the tighter monotonicity constraint *IMC*. Theorem 5 in online Appendix Section 6 shows that *OC* holds at types where *IMC* does not bind and that over intervals where *IMC* does bind, *OC* holds as a weighted average, and thus, effort is distorted down in an average sense. Thus, even for problems where *IMC* does bind, the economics of the situation are maintained.

A Second Sufficient Condition: Single Crossing.—As an alternative sufficient condition for decoupling, say that v satisfies the *single crossing condition* (*SCC*) if for all $\theta > \theta'$, $v(\cdot, \theta)$ single-crosses $v(\cdot, \theta')$ from below. This can again be interpreted as incentives getting stronger as θ increases.[31] Note that *SCC* holds in utility space (that is, for v) if and only if it also holds in cash space. So, as long as the compensation schemes in the menu are observable, *SCC* is observable too.[32] As can be seen from Figure 4, the contracts generated in our numerical example satisfy *SCC* as well as *IMC*.

Theorem 6 in online Appendix Section 6 shows that if a menu is optimal in the relaxed problem $\mathcal{P}_D$ and satisfies *SCC*, then it is feasible (and hence optimal) in $\mathcal{P}$. One potentially fertile ground to utilize *SCC* is with the exponential families of signal distributions. For these (see Chade and Swinkels 2020b), it is easy to show that for any (a, u_0, θ) and (a', u_0', θ'), the solutions to the relaxed moral hazard problem $\mathcal{P}_{MH}$ cross either never or exactly once as a function of x. In any solution to $\mathcal{P}_D$, they must thus cross exactly once, otherwise the type facing the lower contract will deviate. Using Lemma 12 in online Appendix Section 6, we then have that the global property *SCC* holds if and only if the local property *SOC* does.

Common Values.—Some important economic applications with a common value aspect, such as an insurance setting where there is a binary loss, can be analyzed with the tools we developed for the private values case via a reparameterization where effort and ability enter into the signal distribution via an index.[33,34] But, especially once one moves away from the two-outcome case, many problems with common values cannot be reparameterized in this way. Given this, consider an extension of problem $\mathcal{P}$ in which each of $c, f,$ and B can depend on θ. In Proposition 4 in online Appendix Section 6, we provide conditions on primitives that allow us to prove that *SCC* suffices for feasibility of a menu in this more general setting, thus broadening

[31] Neither *SCC* nor *IMC* implies the other. First, *IMC* can hold even though the compensation schemes at different θ cross more than once. When compensation schemes cross only once, they cross in the right direction if and only if *SOC* is satisfied, which is weaker than *IMC*. The two conditions coincide for the two-outcome case.

[32] It is direct from incentive compatibility that if two compensation schemes cross exactly once, then a higher type is associated with the "steeper" compensation scheme.

[33] More generally, assume that for each effort level e and ability θ of the agent, the distribution over the signal is $\hat{f}(x \mid e, \theta) \equiv f(x \mid \eta(e, \theta))$, where η is strictly increasing in e and where the agent has strictly increasing cost of effort $\kappa(e)$. Let $\gamma(a, \theta)$ be given by $\eta(\gamma(a, \theta), \theta) = a$, so that γ is the inverse of η with respect to e given θ. Then, defining $a = \eta(e, \theta)$ and c by $c(a, \theta) = \kappa(\gamma(a, \theta))$, we are back in the private value setting.

[34] In the insurance case there is also a type-dependent participation constraint. Despite this additional difficulty, our tools apply.

the applicability of our results.[35] We see no reason that our numerical techniques cannot be extended to this setting, again, with the verification step involving *SCC* rather than *IMC*.

Optimal Exclusion.—When C is large relative to B, exclusion can easily be relevant in our setting. To analyze this, note first that since high types can imitate low types, any exclusion occurs on an interval of low types $[\underline{\theta}, \theta_c)$. Proposition 5 in online Appendix Section 6 then characterizes the optimal threshold θ_c. Mirroring previous results in the literature, if $h(\underline{\theta})$ is sufficiently small, then there is a strictly positive region of exclusion, which is intuitive, as a small amount of exclusion then destroys surplus on very few agents but reduces rents in a first-order way. Similarly, if B is large enough, and $h(\underline{\theta}) > 0$, then there is no exclusion. We are unaware of any related results at the level of generality of Proposition 5 since our setting subsumes the pure adverse selection problem with quasilinear utility or with risk aversion, and the combined problem.

Solving for optimal exclusion numerically is quite straightforward. For each surplus τ of the highest type, one can solve our pair of differential equations to arrive at a cutoff type below which the solution gives less than the outside option to the agent and then calculate the profits of the mechanism given that types below this cutoff do not participate. We show in online Appendix Section 4 that as one varies τ, one is implicitly searching over all possible cutoff levels. The left-hand panel of Figure 8 shows the surplus schedule for three different exclusion cutoffs. The right-hand panel of Figure 8 shows how the optimal exclusion boundary varies as a function of the intercept of the benefit, b_0.[36] The principal excludes the least under full information, and the most when both frictions are present.

Random Mechanisms.—So far, we have restricted the principal to deterministic menus. In Proposition 6 in online Appendix Section 6, we leverage an idea of Strausz (2006) to show that when $C(\cdot, \cdot, \theta)$ is convex, then if decoupling works, the deterministic menu it generates is in fact optimal even when randomization is allowed. Hence, the restriction to deterministic menus, which is realistic in many settings, is also inconsequential whenever decoupling works.

Principal as Social Planner.—Instead of a profit-maximizing principal, consider a social planner designing a tax code. The code raises and redistributes income and also determines effort incentives. To this end, let h be the distribution of types in an economy with a continuum of agents, and for simplicity, let F be linear in a. The social planner puts weight $1 - \eta$ on the total surplus, $\int Sh$, of the agents and weight η on the profit, $\int (B - C)h$, of the firm. The planner faces a participation constraint that the firm's profits are at least K. Agents have outside option $\bar{u}$, and their types and actions remain hidden.

Under the intuitive assumption that the shadow value of extra utility increases in the type (which we can justify from primitives), we show that in the same average

[35] Intriguingly, an analog to Theorem 1, which centers on *IMC*, seems more difficult. We leave such a generalization and also the exploration of primitives that guarantee *SCC* in the common value case for future work.

[36] Recall that $B(a) = b_0 + 50000 \cdot E[x|a]$.

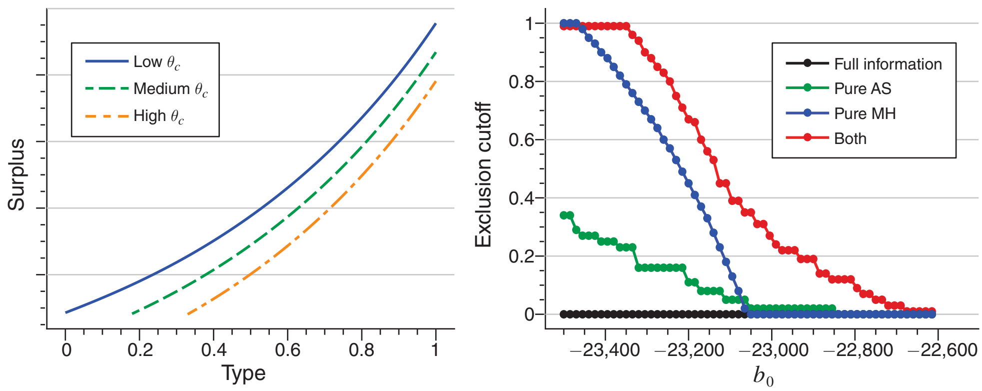

Figure 8. Optimal Exclusion

sense as before, effort is distorted downward and is distorted downward pointwise where both $S(\underline{\theta}) \geq \bar{u}$ and *IMC* is slack. But, unlike the profit-maximizing principal, the social planner distorts effort the most at middle types. This is so as lowering the effort of a middle type lets the principal move utility from the many consumers above this type to the many consumers below this type in an incentive-compatible, budget-balanced, and efficiency-enhancing way.

IX. Concluding Remarks

We study a canonical problem with both moral hazard and adverse selection. Examples include insurance, the social planner's problem of Section VIII, the ubiquitous contracting problem between shareholders and CEOs who both exert effort and know something about their type or project quality, Laffont and Tirole's (1986) procurement problem with a risk-averse agent and standard moral hazard, and extensions of Lewis and Sappington's (2000, 2001) project contracting problem.

We derive necessary conditions for a menu to be feasible. We study a relaxed problem and exhibit how this relaxed problem can be solved through a decoupling procedure that treats moral hazard and screening sequentially. Sufficient for the solution to this problem to be feasible, and hence optimal, is *IMC*—that the recommended action rises fast enough that the agent's marginal cost of effort rises with his type. With linear output, this condition is a characterization.

Decoupling points the way to numerically efficient calculation of a candidate menu in a setting that is intractable from a brute-force perspective. Using our tools, the applied modeler can quickly analyze the behavior of the decoupled solution for their setting and directly verify *IMC*. We provide an illustration using a canonical economic application.

We derive primitives where *IMC* is guaranteed to hold with linear output and when the reservation utility is large. When decoupling works, we have the economically important implication that the optimal menu has the form that the agent, by choice of his announcement of type, is choosing from a menu of compensation schemes of the form given by the relaxed moral hazard problem. Thus, everything that we know from that problem carries over into the setting that also includes adverse selection, and everything that we know from the pure adverse selection problem carries through when there is also moral hazard. There are also interesting implications of the combined information frictions beyond those of each separately.

We show that our results extend naturally to other objective functions such as that of a social planner, with potential applications to optimal taxation. Finally, we study an alternative sufficient condition for feasibility, we begin the generalization of our setup to one with common values, we examine optimal exclusion, and we show that decoupled menus remain optimal even if the principal can randomize.

There are several open problems for future research. We name two: a full analysis of the common values case and the extension to a dynamic setting.

Appendix

A1. *Proofs for Section IV*

LEMMA 1: *The function α is strictly increasing in any solution to $\mathcal{P}$. If α is differentiable and v is twice differentiable in θ, then the second-order necessary conditions are satisfied if and only if $\int v_\theta(x,\theta) f_a(x \mid \alpha(\theta)) dx \geq 0$.*

PROOF:

The objective function of the agent with type θ as a function of his reported type, θ_A, and chosen action, a, is $\int v(x,\theta_A) f(x \mid a) dx - c(a,\theta)$. Its first derivatives, evaluated at the candidate menu, yield IC_{MH} and IC_A, and its second derivatives the Hessian matrix

$$M = \begin{bmatrix} \int v_{\theta\theta}(x,\theta) f(x \mid \alpha(\theta)) dx & \int v_\theta(x,\theta) f_a(x \mid \alpha(\theta)) dx \\ \int v_\theta(x,\theta) f_a(x \mid \alpha(\theta)) dx & \int v(x,\theta) f_{aa}(x \mid \alpha(\theta)) dx - c_{aa}(\alpha(\theta),\theta) \end{bmatrix}.$$

The second-order conditions for optimality by the agent require that M has negative diagonal elements and positive determinant.

Given that feasibility implies that α is increasing and hence almost everywhere differentiable, to show that α is strictly increasing, we need only to rule out that at some point, $\alpha' = 0$. Differentiating the identity IC_A yields

$$\text{(A1)} \qquad \int v_{\theta\theta}(x,\theta) f(x \mid \alpha(\theta)) dx = -\alpha'(\theta) \int v_\theta(x,\theta) f_a(x \mid \alpha(\theta)) dx.$$

Hence, if $\alpha' = 0$, then $\int v_{\theta\theta}(x,\theta) f(x \mid \alpha(\theta)) dx = 0$. Similarly, differentiating IC_{MH} yields

$$\text{(A2)} \qquad \alpha' \Big[\int v(x,\theta) f_{aa}(x \mid \alpha(\theta)) dx - c_{aa}(\alpha(\theta),\theta) \Big]$$

$$= c_{a\theta}(\alpha(\theta),\theta) - \int v_\theta(x,\theta) f_a(x \mid \alpha(\theta)) dx,$$

and so if $\alpha' = 0$, then, $\int v_\theta(x,\theta) f_a(x \mid \alpha(\theta)) dx = c_{a\theta}(\alpha(\theta),\theta) < 0$. But then, $\det M = -\big(c_{a\theta}(\alpha(\theta),\theta)\big)^2 < 0$, violating the second-order necessary conditions.

Finally, let us establish that the second-order necessary conditions hold if and only if *SOC* holds. Since $\alpha' > 0$, it follows from (A1) that $\int v_{\theta\theta} f \leq 0$ if and only if *SOC* holds. Similarly, from (A2), $\int v f_{aa} - c_{aa} \leq 0$ if and only if $c_{a\theta} - \int v_\theta f_a \leq 0$, and since $c_{a\theta} < 0$, it suffices for this that *SOC* holds. To complete the proof, by (A1) and (A2), the determinant of M is

$$\begin{vmatrix} -\alpha' \int v_\theta f_a & \int v_\theta f_a \\ \int v_\theta f_a & \frac{1}{\alpha'}\left(c_{a\theta} - \int v_\theta f_a\right) \end{vmatrix} = -c_{a\theta} \int v_\theta f_a \underset{s}{=} \int v_\theta f_a. \blacksquare$$

A2. *Proofs for Section V*

We begin with a preliminary result. Let the graph of α be $\mathbb{G} \equiv \left\{ \left(\theta, \alpha(\theta)\right) : \theta \in \left[\underline{\theta}, \bar{\theta}\right] \right\}$. Note that in any solution to $\mathcal{P}_D$, $S(\theta_A) + c\left(\alpha(\theta_A), \theta_A\right) - c\left(\alpha(\theta_A), \theta\right)$ is the surplus to type θ of announcing type θ_A and taking action $\alpha(\theta_A)$. Our next lemma says that the agent prefers $\left(\theta, \alpha(\theta)\right)$ to any other announcement-action pair on $\mathbb{G}$.

LEMMA 2: *Let* (α, S) *satisfy* α *increasing and* IC_S*. Then, for each type* θ, $S(\cdot) + c\left(\alpha(\cdot), \cdot\right) - c\left(\alpha(\cdot), \theta\right)$ *is maximized at* θ.

PROOF:

Assume *wlog* that $\theta_A > \theta$. Then using IC_S,

$$\begin{aligned} S(\theta) - S(\theta_A) &= \int_\theta^{\theta_A} c_\theta\left(\alpha(s), s\right) ds \geq \int_\theta^{\theta_A} c_\theta\left(\alpha(\theta_A), s\right) ds \\ &= c\left(\alpha(\theta_A), \theta_A\right) - c\left(\alpha(\theta_A), \theta\right), \end{aligned}$$

where the first equality is by IC_S; the inequality follows since α is increasing and c is strictly submodular in (a, θ); and the second equality uses the fundamental theorem of calculus, which applies because $c_\theta(a, \cdot)$ is a continuous function on the compact set $\left[\underline{\theta}, \bar{\theta}\right]$. Thus, $S(\theta) \geq S(\theta_A) + c\left(\alpha(\theta_A), \theta_A\right) - c\left(\alpha(\theta_A), \theta\right)$ as required. $\blacksquare$

COROLLARY 1: *Let* (α, S) *satisfy* α *increasing and* IC_S*. Then,* $S(\cdot) + c\left(\alpha(\cdot), \cdot\right)$ *is increasing.*

PROOF:

By Lemma 2, for any $\theta' > \theta$, $S(\theta') \geq S(\theta) + c\left(\alpha(\theta), \theta\right) - c\left(\alpha(\theta), \theta'\right)$, and so, since α is increasing, $S(\theta') \geq S(\theta) + c\left(\alpha(\theta), \theta\right) - c\left(\alpha(\theta'), \theta'\right)$, and thus, $S(\cdot) + c\left(\alpha(\cdot), \cdot\right)$ is increasing. $\blacksquare$

PROOF OF THEOREM 1:

Let us show that if (α, S) is feasible in $\mathcal{P}_D$, then (α, v) is feasible in $\mathcal{P}$. So, let (α, S) be feasible in $\mathcal{P}_D$. By IC_S, *IR* holds. From Lemma 2, to show *IC*, it suffices to show that every deviation $(\theta_A, \hat{a})$ is dominated by some deviation in $\mathbb{G}$. We focus on deviations with $\hat{a} \geq \alpha(\theta_A)$ (the other case is similar).

Let θ_T be the true type of the agent. Let us show first that if $\hat{a} > \alpha(\bar{\theta})$, then θ_T prefers the deviation $(\theta_A, \alpha(\bar{\theta}))$ to $(\theta_A, \hat{a})$. To see this, consider any $a \in (\alpha(\bar{\theta}), \hat{a}]$. Then, since $a > \alpha(\theta_A)$,

$$\begin{aligned}\int v(x,\theta_A) f_a(x|a)dx &\leq \int v(x,\theta_A) f_a(x|\alpha(\theta_A))dx \\ &= c_a(\alpha(\theta_A),\theta_A) \\ &\leq c_a(\alpha(\bar{\theta}),\bar{\theta}) \\ &< c_a(a,\bar{\theta}) \\ &\leq c_a(a,\theta_T),\end{aligned}$$

where the first inequality follows by *FOP*, the equality by IC_{MH}, the second inequality by *IMC*, the third by strict convexity of c in a, and the fourth since $\theta_T \leq \bar{\theta}$ and by submodularity of c. Hence, $\int v(x,\theta_A) f_a(x|a)dx < c_a(a,\theta_T)$ for any $a \in (\alpha(\bar{\theta}), \hat{a}]$. Thus, θ_T prefers the deviation $(\theta_A, \alpha(\bar{\theta}))$ to $(\theta_A, \hat{a})$ as claimed.

Given the last step, consider any deviation $(\theta_A, \hat{a})$ with $\hat{a} \in [\alpha(\theta_A), \alpha(\bar{\theta})]$. Since $\hat{a} \in [\alpha(\theta_A), \alpha(\bar{\theta})]$, and since α is continuous, there is $\hat{\theta} \in [\theta_A, \bar{\theta}]$ such that $\hat{a} = \alpha(\hat{\theta})$. We will show that $(\hat{\theta}, \hat{a}) \in \mathbb{G}$ dominates $(\theta_A, \hat{a})$. Consider first moving vertically, for $a \in [\alpha(\theta_A), \hat{a}]$. We have

$$\begin{aligned}\frac{\partial}{\partial a}\left[\int v(x,\theta_A) f(x|a)dx\right] &= \int v(x,\theta_A) f_a(x|a)dx \\ &\leq \int v(x,\theta_A) f_a(x|\alpha(\theta_A))dx \\ &= c_a(\alpha(\theta_A),\theta_A),\end{aligned}$$

appealing to online Appendix Lemma 15, to justify the exchange of differentiation and integration in the first equality, to *FOP* for the inequality, and to IC_{MH} for the second equality. But then, since $\int v(x,\theta_A) f(x|\cdot)dx$ is continuously differentiable, and hence absolutely continuous on $[\alpha(\theta_A), \hat{a}]$, the fundamental theorem of calculus gives us

$$\begin{aligned}\text{(A3)}\quad \int v(x,\theta_A) f(x|\hat{a})dx &= \int v(x,\theta_A) f(x|\alpha(\theta_A))dx \\ &\quad + \int_{\alpha(\theta_A)}^{\hat{a}} \frac{\partial}{\partial a}\left[\int v(x,\theta_A) f(x|a)dx\right] da \\ &\leq \int v(x,\theta_A) f(x|\alpha(\theta_A))dx + [\hat{a} - \alpha(\theta_A)]\, c_a(\alpha(\theta_A),\theta_A).\end{aligned}$$

So, let us consider instead the change in the expected utility of income from moving along the graph from $(\theta_A, \alpha(\theta_A))$ to $(\hat{\theta}, \hat{a})$. Let $\theta \in [\theta_A, \hat{\theta}]$ be a point of differentiability of α. Then, since $v(\cdot,\theta) = \tilde{v}(\cdot,\alpha(\theta),S(\theta),\theta)$, and since $\tilde{v}$ is differentiable, v is differentiable at θ as well. Hence, again appealing to online Appendix Lemma 15,

$$\begin{aligned}
&\frac{\partial}{\partial\theta}\Big[\int v(x,\theta)f\big(x\,|\,\alpha(\theta)\big)dx\Big] \\
&\qquad = \int v_\theta(x,\theta)f\big(x\,|\,\alpha(\theta)\big)dx + \alpha'(\theta)\int v(x,\theta)f_a\big(x\,|\,\alpha(\theta)\big)dx \\
&\qquad = \alpha'(\theta)\int v(x,\theta)f_a\big(x\,|\,\alpha(\theta)\big)dx \\
&\qquad = \alpha'(\theta)c_a\big(\alpha(\theta),\theta\big) \\
&\qquad \geq \alpha'(\theta)c_a\big(\alpha(\theta_A),\theta_A\big),
\end{aligned}$$

where the second equality uses IC_A, the third uses IC_{MH}, and the inequality uses IMC. Thus,

$$\begin{aligned}
\text{(A4)}\quad &\int v\big(x,\hat{\theta}\big)f\Big(x\,|\,\alpha\big(\hat{\theta}\big)\Big)dx \\
&\qquad \geq \int v(x,\theta_A)f\big(x\,|\,\alpha(\theta_A)\big)dx + \int_{\theta_A}^{\hat{\theta}}\frac{\partial}{\partial\theta}\Big[\int v(x,\theta)f\big(x\,|\,\alpha(\theta)\big)dx\Big]d\theta \\
&\qquad \geq \int v(x,\theta_A)f\big(x\,|\,\alpha(\theta_A)\big)dx + \big[\alpha\big(\hat{\theta}\big) - \alpha(\theta_A)\big]c_a\big(\alpha(\theta_A),\theta_A\big),
\end{aligned}$$

where the first inequality follows by Kolmogorov and Fomin (1970, chap. 9), Section 33, Theorem 1, using that $\int v(x,\cdot)f\big(x\,|\,\alpha(\cdot)\big)dx$ (which is equal to $S(\cdot) + c\big(\alpha(\cdot),\cdot\big)$) is increasing.[37]

To complete the proof that (α,v) is feasible in $\mathcal{P}$, note that comparing (A3) and (A4) and recalling that $\alpha\big(\hat{\theta}\big) = \hat{a}$, we have $\int v\big(x,\hat{\theta}\big)f(x\,|\,\hat{a})dx \geq \int v(x,\theta_A)f(x\,|\,\hat{a})dx$, and so θ_T is better-off, given action $\hat{a}$, to announce $\hat{\theta}$ than to announce θ_A. That optimality of (α,S) in $\mathcal{P}_D$ implies optimality of (α,v) in $\mathcal{P}$ follows immediately since $\mathcal{P}_D$ is a relaxation of $\mathcal{P}$. ∎

A3. *Proofs for Section VII*

Prior to proving Theorem 2, we will need three lemmas and an expression for α'. Differentiating OC with respect to θ, recalling that $B_{aa} = 0$ by Assumption 3, and rearranging yields

$$(OC')\qquad \alpha' = \frac{\Big(-c_{a\theta\theta} + c_{a\theta}\frac{h'}{h}\Big)\int_\theta^{\overline{\theta}}\hat{C}_{u_0}h + c_{a\theta}\hat{C}_{u_0}h - \hat{C}_{au_0}c_\theta h + \hat{C}_{a\theta}h}{-\hat{C}_{aa}h + c_{aa\theta}\int_\theta^{\overline{\theta}}\hat{C}_{u_0}h}.$$

Under the assumption that $\hat{C}_{aa}$ is positive, the denominator is strictly negative.

[37] If we knew that $\int v(x,\cdot)f\big(x\,|\,\alpha(\cdot)\big)dx$ was absolutely continuous, then the inequality would be an equality by the fundamental theorem of calculus.

LEMMA 3: *Under Assumption 4, sufficient for IMC is that* $\left[2\lambda + \mu_a + \mu\left(c_{aa\theta}/c_{a\theta}\right)\right]c_{aa} \geq C_{aa}$ *for each* $\left(\alpha(\theta), S(\theta), \theta\right)$.

PROOF:

Recall that α' is given by *OC'* and that *IMC* is the condition that $\alpha' \geq -c_{a\theta}/c_{aa}$ for all θ. Hence, rearranging, *IMC* holds if and only if for all θ,

$$\left(-c_{a\theta\theta}c_{aa} + c_{a\theta}c_{aa}\frac{h'}{h} + c_{a\theta}c_{aa\theta}\right)\int_{\theta}^{\bar{\theta}} C_{u_0}h + \left(c_{a\theta}C_{u_0} - C_{au_0}c_\theta + C_{a\theta}\right)hc_{aa}$$
$$\leq -c_{a\theta}\left(-C_{aa}\right)h$$

since the denominator on the right-hand side of *OC'* is strictly negative. The bracketed term on the left-hand side has the sign of $\left(\partial/\partial\theta\right)\left[\log\left(-c_{a\theta}/hc_{aa}\right)\right]$, which is negative by Assumption 4. Hence, it suffices that

$$\text{(A5)} \qquad \left(c_{a\theta}C_{u_0} - C_{au_0}c_\theta + C_{a\theta}\right)c_{aa} \leq -c_{a\theta}\left(-C_{aa}\right).$$

But it is direct from the envelope theorem that $C_{u_0} = \lambda$ and $C_\theta = \lambda c_\theta + \mu c_{a\theta}$, and hence, since the Lagrange multipliers are continuously differentiable (see Jewitt, Kadan, and Swinkels 2008), $C_{au_0} = \lambda_a$ and $C_{\theta a} = \lambda_a c_\theta + \lambda c_{a\theta} + \mu_a c_{a\theta} + \mu c_{aa\theta}$. Thus,

$$c_{a\theta}C_{u_0} - C_{au_0}c_\theta + C_{a\theta} = 2\lambda c_{a\theta} + \mu_a c_{a\theta} + \mu c_{aa\theta},$$

and the result follows. ∎

The objects λ, μ, and C_{aa} are complicated functions of the primitives. The following lemma helps to break them into more manageable objects. It relies on Lemma 7 of Chade and Swinkels (2020a) characterizing C_{aa}. Recall that ρ is the function that transforms $1/u'$ into u.

LEMMA 4: *If Assumption 4 holds, then a sufficient condition for the second-stage screening solution to satisfy IMC is that for each* θ,

$$\text{(A6)} \qquad \left(\lambda + \mu_a + \mu\frac{c_{aa\theta}}{c_{a\theta}}\right)c_{aa} \geq \mu\left(c_{aaa} - \int vf_{aaa} - \int v_x lF_{aa}\right)$$
$$+ \left[\mu_a^2 \operatorname{var}_\xi(l) - \mu^2 \operatorname{var}_\xi(l_a)\right]\int \rho' f,$$

where for each θ, $v(\cdot, \theta) = v^{MH}\left(\cdot, \alpha(\theta), S(\theta), \theta\right)$, *and* ξ *is the density with kernel* $\rho'\left(\lambda + \mu l\left(\cdot \mid \alpha(\theta)\right)\right)f\left(\cdot \mid \alpha(\theta)\right)$.

PROOF:

From *CS*, Lemma 7,

$$C_{aa} = \lambda c_{aa} + \mu\left(c_{aaa} - \int vf_{aaa} - \int v_x lF_{aa}\right) + \left[\mu_a^2 \operatorname{var}_\xi(l) - \mu^2 \operatorname{var}_\xi(l_a)\right]\int \rho' f,$$

and so, using Lemma 3 and canceling λc_{aa} from each side yields the result. ∎

LEMMA 5: *Let* G *and* $\hat{G}$ *be two CDF.s with finite support and with* $\hat{g}/g$ *continuously differentiable and increasing. Then*,

$$\frac{\mathrm{cov}_G(s^2,s)}{\mathrm{var}_G(s)} \le \frac{\mathrm{cov}_{\hat{G}}(s^2,s)}{\mathrm{var}_{\hat{G}}(s)}.$$

For intuition, note that $\mathrm{cov}_G(s^2,s)/\mathrm{var}_G(s)$ is the slope of the best linear fit to s^2 under G and that $\hat{G}$ moves probability mass rightward from G and hence to where s^2 has a higher slope. See online Appendix Section 5 for the proof.

PROOF OF THEOREM 2:

For the linear case (A6) reduces to

$$\left(\lambda + \mu_a + \mu\frac{c_{aa\theta}}{c_{a\theta}}\right)c_{aa} \ge \mu c_{aaa} + \left[\mu_a^2 \mathrm{var}_\xi(l) - \mu^2 \mathrm{var}_\xi(l_a)\right]\int \rho' f,$$

where, since $c_{aa}/c_{a\theta}$ is increasing in a, and since in the linear case $l_a = -l^2$, it suffices that

$$\lambda c_{aa} \ge -\mu_a c_{aa} + \left[\mu_a^2 \mathrm{var}_\xi(l) - \mu^2 \mathrm{var}_\xi(l^2)\right]\int \rho' f.$$

But, from (14) in the online Appendix (or from *CS*),

$$\begin{aligned}\mu_a &= \frac{1}{\mathrm{var}_\xi(l)}\left[\frac{1}{\int\rho' f}\left(c_{aa} - \int \rho f_{aa}\right) - \mu\,\mathrm{cov}_\xi(l_a,l)\right]\\ &= \frac{1}{\mathrm{var}_\xi(l)}\left[\frac{1}{\int\rho' f}c_{aa} + \mu\,\mathrm{cov}_\xi(l^2,l)\right].\end{aligned}$$

Substituting, it suffices that

$$\begin{aligned}\lambda c_{aa} \ge& -\frac{1}{\mathrm{var}_\xi(l)}\left[\frac{1}{\int\rho' f}c_{aa} + \mu\,\mathrm{cov}_\xi(l^2,l)\right]c_{aa}\\ &+ \left\{\frac{1}{\mathrm{var}_\xi(l)}\left[\frac{1}{\int\rho' f}c_{aa} + \mu\,\mathrm{cov}_\xi(l^2,l)\right]^2 - \mu^2 \mathrm{var}_\xi(l^2)\right\}\int \rho' f,\end{aligned}$$

and so, expanding the bracketed squared term, canceling, and rearranging, it suffices that

$$\lambda c_{aa} \ge c_{aa}\mu\frac{\mathrm{cov}_\xi(l^2,l)}{\mathrm{var}_\xi(l)} + \left(\int\rho' f\right)\frac{\mu^2}{\mathrm{var}_\xi(l)}\left\{\left[\mathrm{cov}_\xi(l^2,l)\right]^2 - \mathrm{var}_\xi(l^2)\mathrm{var}_\xi(l)\right\}.$$

We are thus done if (i) $\mathrm{cov}_\xi(l^2,l) \le 0$ and (ii) $\left[\mathrm{cov}_\xi(l^2,l)\right]^2 - \mathrm{var}_\xi(l^2)\mathrm{var}_\xi(l) \le 0$. Now, $\mathrm{cov}_F(l^2,l) =_s \mathrm{skew}_F(l) \le 0$ since $E_F[l] = 0$ and by assumption. Let $\hat{\xi}$ be the distribution on l generated by ξ and $\hat{f}$ be the distribution on l generated by f.[38]

[38] That is, $\hat{\xi}(l) = \hat{\xi}\left(l^{-1}(l|a)\right)/l_x\left(l^{-1}(l|a)\right)$, and similarly for $\hat{f}$.

Then, since ρ is continuously differentiable and concave, $\hat{f}/\hat{\xi}$ is continuously differentiable and increasing. Result (i) then follows from Lemma 5. To see (ii), note that

$$\left[\operatorname{cov}_\xi\left(l^2,l\right)\right]^2 - \operatorname{var}_\xi\left(l^2\right)\operatorname{var}_\xi(l) = \left(E_\xi\left[\left(l^2 - E_\xi\left[l^2\right]\right)\left(l - E_\xi[l]\right)\right]\right)^2 - E_\xi\left[\left(l^2 - E_\xi\left[l^2\right]\right)^2\right]E_\xi\left[\left(l - E_\xi[l]\right)^2\right],$$

which is negative by the Cauchy-Schwartz inequality. ∎

Let us now turn to the large outside option case. Our results in this section hinge off the following result that builds on *CS*.

LEMMA 6: *Consider the solution to $\mathcal{P}_D$. Under Assumption 6, and evaluated at $\left(\alpha(\theta),S(\theta),\theta\right)$,*

$$\frac{C_a}{\varphi'(\bar{u})c_a},\quad \frac{C_{u_0}}{\varphi'(\bar{u})},\quad \frac{C_{au_0}}{\varphi''(\bar{u})c_a},\quad \frac{C_{a\theta}}{\varphi'(\bar{u})c_{a\theta}},\quad \frac{C_{u_0u_0}}{\varphi''(\bar{u}+c)},\quad \textit{and}\quad \frac{C_{aa}}{\varphi'(\bar{u})c_{aa}}$$

converge to 1 *uniformly as $\bar{u}$ diverges.*

That is, since the agent becomes increasingly risk neutral at higher incomes, the derivatives of C converge in a strong sense to those of the problem without moral hazard.

PROOF:

Note first that for all θ, $S(\theta) - \bar{u}$ is bounded by $-\int c_\theta\left(\alpha(s),s\right)ds$, which is finite independently of $\bar{u}$ and α since the set of actions is bounded. But then, by Assumption 6, $\left[\varphi'\left(S(\theta) + c\left(\alpha(\theta),\theta\right)\right)\right]/\varphi'(\bar{u}) \to 1$ uniformly across all θ, and so, by *CS*, Propositions 3 and 4, $C_a\left(\alpha(\theta),S(\theta),\theta\right)/\left[\varphi'(\bar{u})c_a\left(\alpha(\theta),\theta\right)\right] \to 1$ uniformly across α and θ as $\bar{u}$ diverges, and similarly for each of $C_{u_0}/\varphi'(\bar{u})$, $C_{a\theta}/\left[\varphi'(\bar{u})c_{a\theta}\right]$, and $C_{aa}/\left[\varphi'(\bar{u})c_{aa}\right]$. Similarly, $\varphi'''/\varphi'' \to 0$ since P/A is bounded (see the discussion immediately following Assumption 7 in *CS*), and so $C_{u_0u_0}/\varphi''(\bar{u})$ and $C_{au_0}/\left[\varphi''(\bar{u})c_a\right] \to 1$. ∎

LEMMA 7: *As $\bar{u}$ diverges, the solution α to OC converges uniformly to $\tilde{\alpha}$, the solution to OC_{lim}, and its derivative converges uniformly to*

$$\tilde{\alpha}' = \frac{\left(-c_{a\theta\theta} + c_{a\theta}\frac{h'}{h}\right)(1-H) + 2c_{a\theta}h}{-c_{aa}h + c_{aa\theta}(1-H)}.$$

PROOF:

To see the first claim, start from OC and divide by $\varphi'(\bar{u})$ to arrive at

$$\frac{\beta_1}{\varphi'(\bar{u})} - \frac{C_a}{\varphi'(\bar{u})} + \frac{c_{a\theta}}{h}\int_\theta^{\bar{\theta}}\frac{C_{u_0}}{\varphi'(\bar{u})}h = 0,$$

and then use Lemma 6 to show that the last three terms are arbitrarily close (uniformly with respect to α) to $-c_a + c_{a\theta}(1-H)/h$, as claimed. But then, the solution to the differential equation defining α converges uniformly to the solution to the pointwise solution of α that is inherent in OC_{lim}. To see the second claim, manipulate OC' with $\hat{C} = C$ to arrive at

$$\alpha' = \frac{\left(-c_{a\theta\theta} + c_{a\theta}\frac{h'}{h}\right)\int_{\theta}^{\bar{\theta}} \frac{C_{u_0}}{\varphi'(\bar{u})}h + c_{a\theta}\frac{C_{u_0}}{\varphi'(\bar{u})}h - \frac{C_{au_0}}{\varphi''(\bar{u})}\frac{\varphi''(\bar{u})}{\varphi'(\bar{u})}c_{\theta}h + \frac{C_{a\theta}}{\varphi'(\bar{u})c_{a\theta}}c_{a\theta}h}{-\frac{C_{aa}}{\varphi'(\bar{u})c_{aa}}c_{aa}h + c_{aa\theta}\int_{\theta}^{\bar{\theta}}\frac{C_{u_0}}{\varphi'(\bar{u})}h},$$

and so, by Lemma 6, and using that $\varphi''/\varphi' \to 0$, we have that α' converges uniformly to $\tilde{\alpha}'$. ∎

We will prove Theorem 3 under the following weaker assumption than Assumption 4.

ASSUMPTION 4′: *For each* $a > 0$, $\left[-c_{aa}(a,\cdot)/c_{a\theta}(a,\cdot)\right]\left(h(\cdot)/\left[1 - H(\cdot)\right]\right)$ *is strictly increasing.*[39]

PROOF OF THEOREM 3:

Note that $\tilde{\alpha}' > -c_{a\theta}/c_{aa}$ if and only if

$$\left(-c_{a\theta\theta}c_{aa} + c_{a\theta}c_{aa}\frac{h'}{h} + c_{a\theta}c_{aa\theta}\right)(1 - H) < -c_{a\theta}c_{aa}h,$$

or, using that $c_{a\theta} < 0$, and rearranging,

$$-\frac{-c_{a\theta\theta}}{-c_{a\theta}} + \frac{c_{aa\theta}}{c_{aa}} + \frac{h'}{h} - \frac{-h}{1 - H} > 0,$$

which, since logarithm is an increasing function, is to say that $\left[c_{aa}/(-c_{a\theta})\right] \times \left[h/(1 - H)\right]$ is strictly increasing in θ for each $a > 0$, which is true by Assumption 4′. Thus, $\tilde{\alpha}$ satisfies *IMC* strictly, and so it follows from Lemma 7 that *IMC* holds for sufficiently large $\bar{u}$. ∎

Theorem 3 allows us to conduct comparative statics with respect to shifts in the distribution of types. Say that $\hat{H}$ is *hazard-rate larger* than H if $\hat{h}/(1 - \hat{H})$ is everywhere below $h/(1 - H)$, noting that this implies that the principal faces a more able distribution of types under $\hat{H}$. This subsumes intuitive changes in the distribution of types. Starting from any H with an increasing hazard rate, it is easy to show that if one shifts H to the right and/or stretches it to the right (spreading the quantiles of H), then $\hat{H}$ is hazard-rate larger than H.[40] We have the following comparative statics result.

[39] This is weaker than Assumption 4 since $1 - H$ is strictly decreasing. Indeed, in many settings, strict monotonicity of the hazard rate $h/(1 - H)$ holds, a force in the right direction.

[40] Formally, let $\hat{H}(g(\theta)) = H(\theta)$ for some differentiable g with $g(\theta) \geq \theta$ and $g'(\theta) \geq 1$ for all θ. In turn, $g' \geq 1$ holds if and only if $\hat{H}$ is larger than H in the dispersive order. See Theorem 2.6.6(ii) of Belzunce, Martinez-Riquelme, and Mulero (2016).

PROPOSITION 1 (Comparative Statics of Type Distribution): *Let $\hat{H}$ be hazard-rate larger than H. Then, under the conditions of Theorem 3, and for all $\bar{u}$ sufficiently large, the action schedule in the solution to $\mathcal{P}$ is lower (more distorted) under $\hat{H}$ than under H.*

The proof is immediate. A decrease in the hazard rate decreases the left-hand side of OC_{lim}, and α must fall to restore equality (Maskin and Riley 1984, Proposition 5.3). But since (using Theorem 3) the solution to $\mathcal{P}$ converges to that of OC_{lim}, the same comparative statics hold for any $\bar{u}$ sufficiently large as well. Intuitively, when the hazard rate falls, the efficiency of the action assigned to any given type weighs less heavily compared to the information rents on higher types, thus increasing the optimal distortion.

REFERENCES

Armstrong, Chris, Luzi Hail, and Rachel Xi Zhang. 2022. "Executive Compensation Contracts in the Presence of Adverse Selection." Unpublished.

Baron, David, and David Besanko. 1987. "Monitoring, Moral Hazard, Asymmetric Information, and Risk Sharing in Procurement Contracting." *Rand Journal of Economics* 18 (4): 509–32.

Belzunce, Félix, Carolina Martinez-Riquelme, and Julio Mulero. 2016. *An Introduction to Stochastic Orders.* London: Academic Press.

Bolton, Patrick, and Mathias Dewatripont. 2005. *Contract Theory.* Cambridge, MA: MIT Press.

Castro-Pires, Henrique, Hector Chade, and Jeroen Swinkels. 2024. "Replication data for: Disentangling Moral Hazard and Adverse Selection." American Economic Association [Publisher], Inter-university Consortium for Political and Social Research [Distributor]. https://doi.org/10.3886/E193349V1.

Castro-Pires, Henrique, and Humberto Moreira. 2021. "Limited Liability and Non-Responsiveness in Agency Models." *Games and Economic Behavior* 128: 73–103.

Chade, Hector, and Jeroen Swinkels. 2020a. "The Moral Hazard Problem with High Stakes." *Journal of Economic Theory* 187: 105032.

Chade, Hector, and Jeroen Swinkels. 2020b. "The No-Upward-Crossing Condition and the Moral Hazard Problem." *Theoretical Economics* 15 (2): 446–76.

Edmans, Alex, and Xavier Gabaix. 2016. "Executive Compensation: A Modern Primer." *Journal of Economic Literature* 54 (4): 1232–87.

Fagart, Marie-Cécile. 2002. "Wealth Effects, Moral Hazard and Adverse Selection in a Principal-Agent Model." Unpublished.

Faynzilberg, Peter, and Praveen Kumar. 1997. "Optimal Contracting of Separable Production Technologies." *Games and Economic Behavior* 21 (1-2): 15–39.

Frydman, Carola, and Dirk Jenter. 2010. "CEO Compensation." *Annual Review of Financial Economics* 2 (1): 75–102.

Fudenberg, Drew, and Jean Tirole. 1991. *Game Theory.* Cambridge: Cambridge University Press.

Gottlieb, Daniel, and Humberto Moreira. 2013. "Simultaneous Adverse Selection and Moral Hazard." Unpublished.

Gottlieb, Daniel, and Humberto Moreira. 2022. "Simple Contracts with Adverse Selection and Moral Hazard." *Theoretical Economics* 17 (3): 1357–1401.

Grossman, Sanford, and Oliver Hart. 1983. "An Analysis of the Principal-Agent Problem." *Econometrica* 51 (1): 7–45.

Guesnerie, Roger, and Jean-Jacques Laffont. 1984. "A Complete Solution to a Class of Principal-Agent Problems with an Application to the Control of a Self-Managed Firm." *Journal of Public Economics* 25 (3): 329–69.

Halac, Marina, Navin Kartik, and Qingmin Liu. 2016. "Optimal Contracts for Experimentation." *Review of Economic Studies* 83 (3): 1040–91.

Holmström, Bengt. 1979. "Moral Hazard and Observability." *Bell Journal of Economics* 10 (1): 74–91.

Jewitt, Ian. 1988. "Justifying the First-Order Approach to Principal-Agent Problems." *Econometrica* 56 (5): 1177–90.

Jewitt, Ian, Ohad Kadan, and Jeroen Swinkels. 2008. "Moral Hazard with Bounded Payments." *Journal of Economic Theory* 143 (1): 59–82.

Kaplan, Steven N., and Per Stromberg. 2001. "Venture Capitals as Principals: Contracting, Screening, and Monitoring." *American Economic Review* 91 (2): 426–30.

Kirkegaard, Rene. 2017. "Moral Hazard and the Spanning Condition without the First-Order Approach." *Games and Economic Behavior* 102: 373–87.

Kolmogorov, A., and S. Fomin. 1970. *Introductory Real Analysis.* Mineola, NY: Dover.

Laffont, Jean-Jacques, and David Martimort. 2001. *The Theory of Incentives: The Principal-Agent Model.* Princeton, NJ: Princeton University Press.

Laffont, Jean-Jacques, and Jean Tirole. 1986. "Using Cost Observations to Regulate Firms." *Journal of Political Economy* 94 (3): 614–41.

Laffont, Jean-Jacques, and Jean Tirole. 1993. *A Theory of Incentives in Procurement and Regulation.* Cambridge, MA: MIT Press.

Lewis, Tracy, and David Sappington. 2000. "Contracting with Wealth-Constrained Agents." *International Economic Review* 41 (3): 743–67.

Lewis, Tracy, and David Sappington. 2001. "Optimal Contracting with Private Knowledge of Wealth and Ability." *Review of Economic Studies* 68 (1): 21–44.

LiCalzi, Marco, and Sandrine Spaeter. 2003. "Distributions for the First-Order Approach to Principal-Agent Problems." *Economic Theory* 21 (1): 167–73.

Manove, Michael, A. Jorge Padilla, and Marco Pagano. 2001. "Collateral versus Project Screening: A Model of Lazy Banks." *RAND Journal of Economics* 32 (4): 726–44.

Maskin, Eric, and John Riley. 1984. "Monopoly with Incomplete Information." *RAND Journal of Economics* 15 (2): 171–96.

Milgrom, Paul, and Ilya Segal. 2002. "Envelope Theorems for Arbitrary Choice Sets." *Econometrica* 70 (2): 583–601.

Mirrlees, James. 1975. "On Moral Hazard and the Theory of Unobservable Behavior."

Moroni, Sofia, and Jeroen Swinkels. 2014. "Existence and Non-Existence in the Moral Hazard Problem." *Journal of Economic Theory* 150: 668–82.

Murphy, Kevin. 2013. "Executive Compensation: Where We Are, and How We Got There." In *Handbook of the Economics of Finance*, Vol. 2, edited by George M. Constantinides, Milton Harris, and Rene M. Stulz, 211–356. Amsterdam: Elsevier.

Myerson, Roger. 1981. "Optimal Auction Design." *Mathematics of Operations Research* 6 (1): 58–73.

Myerson, Roger. 1982. "Optimal Coordination Mechanisms in Generalized Principal-Agent Problems." *Journal of Mathematical Economics* 10 (1): 67–81.

Rogerson, William. 1985. "The First-Order Approach to Principal-Agents Problems." *Econometrica* 53 (6): 1357–67.

Strausz, Roland. 2006. "Deterministic versus Stochastic Mechanisms in Principal-Agent Models." *Journal of Economic Theory* 128 (1): 306–14.

Strulovici, Bruno. 2011. "Contracts, Information Persistence, and Renegotiation." Unpublished.

Williams, Noah. 2015. "A Solvable Continuous Time Principal Agent Model." *Journal of Economic Theory* 159: 989–1015.